THE DISCREPANCY OF $\{n\alpha+\beta\}$ AT A FIXED RATIONAL STARTING POINT: KESTEN'S QUESTION AND CAUCHY LAWS WITH ARITHMETIC SCALES

VERIFIABLE MATHEMATICAL REASONING CORPUS

ABSTRACT. For $\alpha,\beta\in\mathbb{R}$ and $0<r<1$ let $D_N(\alpha,\beta;r)=\#\{1\le n\le N:\{n\alpha+\beta\}<r\}-Nr$ be the local discrepancy of the sequence $(\{n\alpha+\beta\})_{n\ge1}$. Kesten proved in 1960 that $D_N/\log N$ converges in distribution to a Cauchy law when (α,β) is uniformly distributed on the unit square, and raised the question whether such a limit law holds for a *fixed* starting point β; this question is recorded as open in the recent literature. We prove that for every rational starting point β and every rational $r\in(0,1)$ the normalised discrepancy $D_N(\alpha,\beta;r)/\log N$, with α uniformly distributed on $(0,1)$, converges to a centred Cauchy law, and we determine its scale: it is $\sigma(\beta,r)=\frac{6}{\pi}\,\mathbb{E}\,F(\{z_1c\},\{z_2c\})$, where $z_1=\{-\beta\}$, $z_2=\{r-\beta\}$, $F(x,y)=\int_0^1|\tau(t,y)-\tau(t,x)|\,dt$ with the Green function $\tau(t,z)=\min(t,z)(1-\max(t,z))$, and c is the second coordinate of a uniformly distributed primitive residue vector modulo the common denominator of z_1,z_2. In particular $\sigma(0,r)=1/(2\pi)$ for every rational r, whereas Kesten's scale for a random starting point is $1/(3\pi)$; other values include $1/(4\pi)$, $3/(8\pi)$ and, for some pairs such as $(\beta,r)=(\frac13,\frac12)$, Kesten's value $1/(3\pi)$ itself. So a Cauchy law survives at every fixed rational starting point, but in general with a scale different from Kesten's. The proof combines three ingredients: an exact reciprocity formula expressing D_N as an alternating Birkhoff sum along the orbit of $(\alpha,\{N\alpha\},z)$ under the skew product $(x,y,z)\mapsto(\{1/x\},\{-y/x\},\{-z/x\})$; the conditional law of the fibre variable $\{N_k\alpha_k\}$ given a filtration by renormalisation cells, obtained from an exactly closed two-piece transfer; and, for the interval endpoints, a finite extension of a Markov refinement of the Gauss map whose exponential mixing we prove by a coupling argument. The value $1/(2\pi)$ at $\beta=0$ comes from the identity $\sum B_2(c/Q)=0$, the sum running over primitive residue vectors (b,c) modulo Q.

1. INTRODUCTION

For real t let $\{t\}=t-\lfloor t\rfloor$. For $\alpha,\beta\in\mathbb{R}$, $0<r<1$ and $N\ge1$ consider the local discrepancy

$$D_N(\alpha,\beta;r)=\#\{1\le n\le N:\ \{n\alpha+\beta\}<r\}-Nr,\qquad D_N(\alpha;r):=D_N(\alpha,0;r).$$

For fixed irrational α the sequence D_N is unbounded unless $r\in\mathbb{Z}+\alpha\mathbb{Z}$ (Hecke, Ostrowski, Kesten), and it oscillates on the scale $\log N$. Kesten [9] discovered that this oscillation has a universal statistical description: if (α,β) is uniformly distributed on $[0,1]^2$, then $D_N(\alpha,\beta;r)/\log N$ converges in distribution to a centred Cauchy law. Kesten expressed the scale by a complicated formula which appeared to depend on the arithmetic nature of r; Borda [3] showed that it equals $1/(3\pi)$ for all $r\in(0,1)$, and that the analogous law for the sawtooth function $\{x\}-\frac12$ has scale $1/(4\pi)$. In Kesten's theorem the randomness comes from *two* parameters. Whether a limit law persists for a *fixed* starting point β, so that all the randomness has to come from α, "was already raised by Kesten, and remains open" [3]; it is also recorded as open in [2]. For the sawtooth function and $\beta=0$ this is Erdős Problem #1002, for which proofs of a Cauchy law with scale $1/(2\pi)$ have recently been claimed by Kwon and by Wang [12], and given independently in [6]. For the indicator of an interval, i.e. for the discrepancy itself, the limit law at a fixed starting point has not been determined. Borda [3] determined, for $\beta=0$ and rational r, the limit laws of $\max_{N<M}D_N$ and $\min_{N<M}D_N$, and asked whether this extends to irrational r; these maximal

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sums only involve α, whereas D_N at a deterministic time N depends on the interplay of N with the continued fraction of α. If the time is randomised as well (N uniform in $\{1,\dots,M\}$), limit laws at a fixed starting point are available for the sawtooth function through the work of Dolgopyat and Sarig [5] (see the discussion in [2]); the present paper concerns a deterministic time N.

The result. For an integer $Q \ge 1$ let $\mathrm{Prim}_Q = \{(b,c) \in (\mathbb{Z}/Q\mathbb{Z})^2 : \gcd(b,c,Q) = 1\}$ be the set of primitive residue vectors, and let c_Q denote the second coordinate of a uniformly distributed element of Prim_Q. For $t, z \in [0,1]$ let

$$\tau(t,z) = \min(t,z)\,\bigl(1 - \max(t,z)\bigr)$$

(the Green function of $-d^2/dt^2$ on $[0,1]$ with Dirichlet boundary conditions), and for $x, y \in [0,1)$ let

$$F(x,y) = \int_0^1 \bigl|\tau(t,y) - \tau(t,x)\bigr|\,dt = \frac{|y-x|\,\bigl(\min(x,y)^2 + (1-\max(x,y))^2\bigr)}{2\,(1-|y-x|)}.$$

(The closed form is elementary: for $x < y$ the integrand is $t(y-x)$ on $[0,x]$, $(y-x)(1-t)$ on $[y,1]$ and $|t(1-y+x)-x|$ on $[x,y]$.)

Theorem 1.1. *Let $\beta \in \mathbb{Q}$ and $r \in \mathbb{Q} \cap (0,1)$. Put $z_1 = \{-\beta\}$, $z_2 = \{r-\beta\}$, let $Q \ge 2$ be the least common denominator of z_1 and z_2, and*

$$\sigma(\beta, r) = \frac{6}{\pi}\,\mathbb{E}\,F\bigl(\{z_1 c_Q\}, \{z_2 c_Q\}\bigr) \;> 0.$$

Then for every $c \in \mathbb{R}$

$$\lim_{N\to\infty} \Bigl|\Bigl\{\alpha \in (0,1) :\ \frac{D_N(\alpha,\beta;r)}{\log N} \le c\Bigr\}\Bigr| = \frac{1}{2} + \frac{1}{\pi}\arctan\frac{c}{\sigma(\beta,r)};$$

that is, if α is uniformly distributed on $(0,1)$ then $D_N(\alpha,\beta;r)/\log N$ converges in distribution to the centred Cauchy law with scale $\sigma(\beta,r)$ (characteristic function $e^{-\sigma(\beta,r)|\theta|}$).

The quantity $\sigma(\beta,r)$ is a finite average: $\mathbb{P}(c_Q = c) = \varphi(d)/(dQ\prod_{p\mid Q}(1-p^{-2}))$ with $d = \gcd(c,Q)$, so $\pi\sigma(\beta,r)$ is a rational number which can be computed exactly. Its positivity is clear, since $c_Q = 1$ has positive probability and $\{z_1\} \ne \{z_2\}$.

Corollary 1.2. *For every rational $r \in (0,1)$, $D_N(\alpha;r)/\log N$ converges in distribution to the Cauchy law with scale $1/(2\pi)$, i.e.*

$$\lim_{N\to\infty} \Bigl|\Bigl\{\alpha \in (0,1) :\ \#\{n \le N : \{n\alpha\} < r\} - Nr \le c\log N\Bigr\}\Bigr| = \frac{1}{2} + \frac{1}{\pi}\arctan(2\pi c) \qquad (c \in \mathbb{R}).$$

Proof. For $\beta = 0$ we have $z_1 = 0$, $z_2 = r$, and Q is the denominator of r. Since $F(0,y) = \int_0^1 \tau(t,y)\,dt = y(1-y)/2$, and $\mathbb{E}\bigl[\{rc_Q\}(1-\{rc_Q\})\bigr] = \frac{1}{6}$ by Lemma 3.8 below (applied with $P = rQ$, which is coprime to Q), we get $\sigma(0,r) = \frac{6}{\pi}\cdot\frac{1}{12} = \frac{1}{2\pi}$. □

The scale genuinely depends on the starting point. Exact values of $\pi\sigma(\beta,r)$ for some pairs are:

(β,r)	$(0,r)$	$(\frac14,\frac12)$	$(\frac15,\frac25)$	$(\frac23,\frac13)$	$(\frac17,\frac37)$	$(\frac13,\frac12)$	$(\frac12,\frac13)$	$(\frac{7}{10},\frac{3}{10})$
$\pi\sigma(\beta,r)$	$\frac12$	$\frac14$	$\frac14$	$\frac14$	$\frac{3}{10}$	$\frac13$	$\frac38$	$\frac{499}{1512}$

Thus the answer to Kesten's question, for rational parameters, is: *a Cauchy limit law holds at every fixed rational starting point, but its scale is an arithmetic function of (β,r) which in general differs from Kesten's value $1/(3\pi)$.* For $\beta = 0$ it is $1/(2\pi)$ for every r, so Kesten's limit law does not hold at the starting point 0; for some other pairs, for instance $(\frac13,\frac12)$, $(\frac17,\frac12)$ or $(\frac14,\frac34)$, the scale happens to equal $1/(3\pi)$. Among all pairs with denominators at most 12 the values of $\pi\sigma(\beta,r)$ lie in $[\frac14,\frac12]$. The values in the table are confirmed by Monte Carlo simulation in Section 11.

Remark 1.3 (Consistency with Kesten's theorem)**.** For a uniformly distributed starting point the pair of endpoints becomes $(Z, \{Z+\delta\})$ with Z uniform and $\delta = \{rc_Q\}$, and a direct computation gives $\int_0^1 F(z, \{z+\delta\})\, dz = \delta(1-\delta)/3$. Since $\mathbb{E}[\delta(1-\delta)] = \frac{1}{6}$ (Lemma 3.8), the same recipe produces the scale $\frac{6}{\pi} \cdot \frac{1}{18} = \frac{1}{3\pi}$, which is Kesten's constant as computed by Borda. Likewise, if $(z_1 c, z_2 c)$ is replaced by a pair of independent uniform variables, $\mathbb{E}F = \frac{1}{18}$.

Remark 1.4 (Irrational parameters)**.** Our proof uses the rationality of β and r in one place only: it makes the orbits of the interval endpoints under the renormalisation a *finite* extension of a Markov refinement of the Gauss map (Section 3). Everything else applies verbatim to arbitrary parameters. The numerical evidence of Section 11 supports the following statement, which we do not prove.

Conjecture 1.5. *For all $\beta \in \mathbb{R}$ and $r \in (0,1)$, $D_N(\alpha, \beta; r)/\log N$ converges to a centred Cauchy law. Its scale is $\frac{6}{\pi}\mathbb{E}F(X_1, X_2)$, where (X_1, X_2) is the limit in distribution of $(\{z_1 q_K\}, \{z_2 q_K\})$, q_K being the denominator of the K-th convergent of α, with α uniform and K uniform in $\{1, \dots, k\}$, $k \to \infty$. In particular the scale is $1/(2\pi)$ for $\beta \in \mathbb{Z}$ and every r, and $1/(3\pi)$ whenever $1, \beta, r$ are linearly independent over $\mathbb{Q}$.*

The method. Our starting point is an exact reciprocity formula for the local discrepancy (Proposition 2.4): with $N_0 = N$, $N_{k+1} = \lfloor N_k \alpha_k \rfloor$, $y_k = \{N_k \alpha_k\}$ and $z_{k+1} = \{-z_k/\alpha_k\}$,

$$D_N(\alpha; z) = \sum_{k=0}^{J-1} (-1)^k \, \ell(\alpha_k, y_k, z_k), \qquad \ell(x, y, z) = \frac{\tau(y, z)}{x} + \kappa(x, y, z), \quad |\kappa| \le 1,$$

where α_k are the Gauss iterates of α and $J \approx \frac{12 \log 2}{\pi^2} \log N$. Thus D_N is an alternating Birkhoff sum along an orbit of the skew product $(x, y, z) \mapsto (\{1/x\}, \{-y/x\}, \{-z/x\})$, and $D_N(\alpha, \beta; r) = D_N(\alpha; z_2) - D_N(\alpha; z_1)$ is a difference of two such sums driven by the same (α_k, y_k). The large terms are $a_{k+1}(\tau(y_k, z_{2,k}) - \tau(y_k, z_{1,k}))$ with a_{k+1} the partial quotients. The proof has to control three kinds of randomness.

(i) The digits. Large partial quotients produce, after normalisation by $\log N$, a Poisson process of jumps with intensity $\frac{c_0}{\log 2}\, dx/x^2$, $c_0 = \frac{12 \log 2}{\pi^2}$; this is where the Cauchy law comes from.

(ii) The fibre variable y_k, which encodes the position of N relative to the continued fraction of α. Its conditional law given the past is not uniform. We use the description of this conditional law obtained in [6]: on a filtration by "renormalisation cells", the fibre map transforms piecewise constant densities with pieces $[0, 1-x)$, $[1-x, 1)$ into densities of the same kind, the mass of the upper piece following a scalar affine recursion with a uniform two-step contraction. We reproduce the statements and proofs (Sections 4–5) to keep the paper self-contained.

(iii) The endpoint variables $z_{i,k}$, which do not depend on N. For a rational endpoint $z = P/Q$ we show that $z_k = \{u_k + v_k \alpha_k\}$, where $u_k = \{(-1)^k P q_k/Q\}$ and v_k is one of finitely many rationals (Lemma 3.1). The pair (u_k, v_k) is driven by the digits *and* by the position of α_k relative to finitely many rational thresholds; refining the Gauss partition at these thresholds gives a Markov partition for which (u_k, v_k) is adapted. We prove exponential mixing of the resulting finite extension by a coupling argument which uses large partial quotients to reset the integer parts of v_k and the fact that the matrices $\left(\begin{smallmatrix} 0 & -1 \\ -1 & -c \end{smallmatrix}\right)$, $c \in \mathbb{Z}/Q$, generate $\mathrm{SL}_2(\mathbb{Z}/Q\mathbb{Z})$ up to sign of the determinant (Section 3). The dynamics of (u_k, v_k) does not depend on the parity of k, so even and odd levels have the same limiting statistics; this is what centres the limit law. The stationary law of u_k is that of $\{Pc_Q/Q\}$, and the elementary identity

$$\sum_{(b,c) \in \mathrm{Prim}_Q} B_2\Big(\Big\{\frac{c}{Q}\Big\}\Big) = 0 \qquad (Q \ge 2), \qquad B_2(x) = x^2 - x + \tfrac{1}{6},$$

shows that the scale at $\beta = 0$ does not depend on r.

The rest of the argument follows [6]: the bounded part of the sum is controlled by a martingale decomposition along the cell filtration, now with discontinuous summands, and the residual drift by the mixing of the extended chain; the large partial quotients are handled by a conditional characteristic function argument and a decoupling of the near, transition and far future.

Organisation. Section 2 proves the reciprocity expansion. Section 3 studies the endpoint orbits for rational endpoints and proves the mixing of the extended chain. Sections 4–5 describe the conditional law of y_k. Section 6 collects facts on continued fractions. Sections 7–9 carry out the decomposition, the bounded part and the large partial quotients, and Section 10 completes the proof. Section 11 reports exact and Monte Carlo checks.

Notation. λ is Lebesgue measure on $(0,1)$, α is λ-distributed, and $\mathbb{P},\mathbb{E}$ refer to λ. For $x\in\mathbb{R}$, $\|x\|$ is the distance to the nearest integer; for $u,u'\in\mathbb{R}/\mathbb{Z}$ we write $\|u-u'\|$ for the circle distance. Constants $C, C_0, C_1, \dots$ are absolute unless their dependence on Q is indicated; $c_0=\frac{12\log 2}{\pi^2}$.

2. The reciprocity expansion

Throughout, $\alpha\in(0,1)$ is irrational, $\alpha=[0;a_1,a_2,\dots]$, $G(x)=\{1/x\}$ is the Gauss map, $\alpha_k=G^k\alpha$ (so $a_{k+1}=\lfloor 1/\alpha_k\rfloor$ and $1/\alpha_k=a_{k+1}+\alpha_{k+1}$), and p_k/q_k are the convergents, $q_{-1}=0$, $q_0=1$, $q_{k+1}=a_{k+1}q_k+q_{k-1}$, and similarly for p_k with $p_{-1}=1$, $p_0=0$. We use the classical facts

$$|q_k\alpha-p_k|=\alpha_0\alpha_1\cdots\alpha_k=:|\theta_k|,\qquad \frac{1}{2q_{k+1}}<|\theta_k|<\frac{1}{q_{k+1}},\qquad \alpha_k\alpha_{k+1}<\frac12, \tag{2.1}$$

and, on the cylinder $\{a_1,\dots,a_k \text{ fixed}\}$, $\alpha=\frac{p_k+p_{k-1}\alpha_k}{q_k+q_{k-1}\alpha_k}$.

The fibre variable of N. Fix $N\ge 1$ and define recursively $N_0=N$ and, for $k\ge 0$,

$$Y_k=N_k\alpha_k,\qquad N_{k+1}=\lfloor Y_k\rfloor,\qquad y_k=\{Y_k\},\qquad J=\min\{k:\ N_k=0\}.$$

Since $N_{k+1}\le N_k\alpha_k<N_k$ as long as $N_k\ge 1$, J is finite; $y_k=0$ for $k\ge J$.

Lemma 2.1. *$y_{k+1}=\{-y_k/\alpha_k\}$ for all $k\ge 0$.*

Proof. $N_{k+1}=Y_k-y_k$ gives $Y_{k+1}=N_{k+1}(1/\alpha_k-a_{k+1})=N_k-a_{k+1}N_{k+1}-y_k/\alpha_k$, and $N_k-a_{k+1}N_{k+1}\in\mathbb{Z}$. □

Lemma 2.2 (Size of the renormalised times)**.** *For all $k\ge 0$,*

$$Y_k=N|\theta_k|-\sum_{i<k} y_i\,\alpha_{i+1}\cdots\alpha_k,\qquad \text{so}\qquad N|\theta_k|-3<Y_k\le N|\theta_k|. \tag{2.2}$$

Consequently $\frac{N}{2q_{k+1}}-4<N_{k+1}<\frac{N}{q_{k+1}}$. If $m=\max\{j:q_j\le N\}$, then $m-8\le J\le m+1$.

Proof. $Y_0=N\alpha=N|\theta_0|$ and $Y_k=(Y_{k-1}-y_{k-1})\alpha_k$, which gives (2.2) by induction. A product of t consecutive α_j is at most $2^{-\lfloor t/2\rfloor}$ by (2.1), so $\sum_{i<k}\alpha_{i+1}\cdots\alpha_k<1+2\sum_{t\ge1}2^{-t}=3$. The bounds on $N_{k+1}=\lfloor Y_k\rfloor$ follow from (2.1). If $q_{k+1}>N$ then $N_{k+1}=0$, so $J\le m+1$. If $q_{k+1}\le N/10$ then $N_{k+1}\ge 1$, so $J\ge k+2$; since $q_{j+2}\ge 2q_j$, at most 8 indices j satisfy $N/10<q_j\le N$, whence $J\ge m-8$. □

The endpoint orbit and the level function. For $z \in [0,1)$ put $z_0 = z$ and $z_{k+1} = \{-z_k/\alpha_k\}$. For $x \in (0,1)$ and $y, z \in [0,1)$ write $y' = \{-y/x\}$, $z' = \{-z/x\}$ and define

$$\kappa(x,y,z) = \begin{cases} -(1-z') & \text{if } y \geq z, \\ -(1-y') + \mathbf{1}\{y' < z'\} & \text{if } y < z, \end{cases} \qquad \ell(x,y,z) = \frac{\tau(y,z)}{x} + \kappa(x,y,z),$$

with $\tau(y,z) = \min(y,z)(1-\max(y,z))$ as in the introduction. Note $|\kappa| \leq 1$, $0 \leq \tau \leq \frac{1}{4}$, $\tau(y,0) = 0$, and τ is 1-Lipschitz in each variable on the circle $\mathbb{R}/\mathbb{Z}$ (it is continuous at $y \in \{0,1\}$ and at $z \in \{0,1\}$, where it vanishes). Write $D_N(\alpha; z) = D_N(\alpha, 0; z)$ and $D_0(\alpha; z) = 0$.

Lemma 2.3 (Local reciprocity)**.** *Let $z \in (0,1)$ with $z \notin \mathbb{Z} + \alpha\mathbb{Z}$. Then for every $N \geq 1$,*

$$D_N(\alpha; z) = \ell\big(\alpha, \{N\alpha\}, z\big) - D_{\lfloor N\alpha \rfloor}\big(G\alpha; \{-z/\alpha\}\big).$$

Moreover $\{-z/\alpha\} \notin \mathbb{Z} + G\alpha\,\mathbb{Z}$.

Proof. Put $M = \lfloor N\alpha \rfloor$, $y = \{N\alpha\}$, $\alpha_1 = G\alpha$, $\rho = \{z/\alpha\}$ and $z_1 = \{-z/\alpha\}$, so $z_1 = 1 - \rho$ (as $z/\alpha \notin \mathbb{Z}$), and $y_1 = \{-y/\alpha\}$. Let $C_N(\alpha; z) = \#\{1 \leq n \leq N : \{n\alpha\} < z\}$. Every $n \in [1, N]$ has $m := \lfloor n\alpha \rfloor \in [0, M]$, and $\{n\alpha\} < z$ iff $m/\alpha \leq n < (m+z)/\alpha$. Hence $C_N = \sum_{m=0}^{M} \#A_m$ with $A_m = \{n \in \mathbb{Z} \cap [1, N] :\ m/\alpha \leq n < (m+z)/\alpha\}$. We use that an interval $[x, x+h)$ with $h > 0$ contains exactly $\lfloor h \rfloor + \mathbf{1}\{\{-x\} < \{h\}\}$ integers. The hypothesis $z \notin \mathbb{Z} + \alpha\mathbb{Z}$ excludes all the equalities between fractional parts that occur below: $\{-m/\alpha\} = \rho$ would give $m + z \in \alpha\mathbb{Z}$, and $y_1 = z_1$ would give $z - y = z - N\alpha + M \in \alpha\mathbb{Z}$.

Case $M = 0$. Then $y = N\alpha$ and $C_N = \#\{1 \leq n \leq N : n < z/\alpha\} = \min(N, \lfloor z/\alpha \rfloor)$. If $y \geq z$, then $C_N = \lfloor z/\alpha \rfloor = z/\alpha - 1 + z_1$ and $D_N = z(1-y)/\alpha - (1 - z_1) = \ell(\alpha, y, z)$. If $y < z$, then $C_N = N$, $D_N = N(1-z) = y(1-z)/\alpha$, and $y_1 = \{-N\} = 0 < z_1$, so again $D_N = \ell(\alpha, y, z)$.

Case $M \geq 1$. For $1 \leq m \leq M-1$ we have $(m+z)/\alpha < M/\alpha \leq N$, so $\#A_m = \lfloor z/\alpha \rfloor + \mathbf{1}\{\{-m\alpha_1\} < \rho\}$, because $\{-m/\alpha\} = \{-m\alpha_1\}$. As $\{-m\alpha_1\} = 1 - \{m\alpha_1\}$, this indicator equals $1 - \mathbf{1}\{\{m\alpha_1\} < z_1\}$. Further $\#A_0 = \lceil z/\alpha \rceil - 1 = \lfloor z/\alpha \rfloor$. Since $\{M\alpha_1\} = \{M/\alpha\} = \{N - y/\alpha\} = y_1$, we obtain

$$\sum_{m=0}^{M-1} \#A_m = M\lfloor z/\alpha \rfloor + (M-1) - C_{M-1}(\alpha_1; z_1) = M\lfloor z/\alpha \rfloor + M - 1 - C_M(\alpha_1; z_1) + \mathbf{1}\{y_1 < z_1\}.$$

For $m = M$ note $N = (M+y)/\alpha$ and $\{-M/\alpha\} = \{y/\alpha\} = 1 - y_1$ (here $y/\alpha \notin \mathbb{Z}$, since $M \geq 1$ and α is irrational). If $y \geq z$, then $(M+z)/\alpha \leq N$ and $\#A_M = \lfloor z/\alpha \rfloor + \mathbf{1}\{1 - y_1 < \rho\} = \lfloor z/\alpha \rfloor + \mathbf{1}\{y_1 > z_1\}$. If $y < z$, then $\#A_M = N - \lceil M/\alpha \rceil + 1 = N - M/\alpha - (1 - y_1) + 1 = y/\alpha + y_1$. Now $C_N = \sum_{m \leq M} \#A_m$, $C_M(\alpha_1; z_1) = D_M(\alpha_1; z_1) + Mz_1$ and $M(\lfloor z/\alpha \rfloor + 1 - z_1) - Nz = Mz/\alpha - Nz = -yz/\alpha$ give

$$D_N(\alpha; z) = -D_M(\alpha_1; z_1) - \frac{yz}{\alpha} - 1 + \mathbf{1}\{y_1 < z_1\} + \#A_M.$$

If $y \geq z$ the last four terms add up to $\lfloor z/\alpha \rfloor - yz/\alpha = z(1-y)/\alpha - (1 - z_1)$; if $y < z$ they add up to $y(1-z)/\alpha - (1 - y_1) + \mathbf{1}\{y_1 < z_1\}$. In both cases this is $\ell(\alpha, y, z)$.

Finally, if $z_1 = j + k\alpha_1$ with $j, k \in \mathbb{Z}$, then $-z/\alpha \in \mathbb{Z} + k(1/\alpha - a_1)$ and $z \in \mathbb{Z} + \alpha\mathbb{Z}$, a contradiction. □

Proposition 2.4 (Alternating expansion)**.** *Let $z \in (0,1) \setminus (\mathbb{Z} + \alpha\mathbb{Z})$, in particular any rational $z \in (0,1)$. For every $N \geq 1$,*

$$D_N(\alpha; z) = \sum_{k=0}^{J-1} (-1)^k \ell_k(z), \qquad \ell_k(z) := \ell(\alpha_k, y_k, z_k). \tag{2.3}$$

Moreover $\ell_k(z) = (a_{k+1} + \alpha_{k+1})\tau(y_k, z_k) + \kappa_k(z)$ with $\kappa_k(z) = \kappa(\alpha_k, y_k, z_k) \in [-1, 1]$, and $|\ell_k(z)| \leq a_{k+1} + 1$. In $\kappa_k(z)$ the quantities y', z' are y_{k+1} and z_{k+1}.

Proof. Apply Lemma 2.3 to (α_k, N_k, z_k) for $k = 0, \dots, J-1$; the hypothesis propagates by the last statement of the lemma, and the recursion stops because $D_0 = 0$. The identification $y' = y_{k+1}$ is Lemma 2.1. Finally $|\ell_k| \le (a_{k+1}+1)/4 + 1 \le a_{k+1} + 1$. □

Thus $D_N(\alpha; z)$ is the alternating Birkhoff sum of ℓ along the orbit of $(\alpha, \{N\alpha\}, z)$ under the skew product $(x, y, z) \mapsto (\{1/x\}, \{-y/x\}, \{-z/x\})$, stopped at the absorption time J of the second coordinate. For general rational starting points we use the following reduction.

Lemma 2.5. *Let $\beta \in \mathbb{R}$, $0 < r < 1$, $z_1 = \{-\beta\}$, $z_2 = \{r - \beta\}$. Then $D_N(\alpha, \beta; r) = D_N(\alpha; z_2) - D_N(\alpha; z_1)$ for all N and all α.*

Proof. $\{n\alpha + \beta\} < r$ iff $\{n\alpha\}$ lies in the arc of length r starting at z_1. If $z_1 < z_2$ this arc is $[z_1, z_2)$ and the count is $C_N(\alpha; z_2) - C_N(\alpha; z_1)$, while $N(z_2 - z_1) = Nr$. If $z_2 < z_1$ (in particular if $z_2 = 0$) the arc is $[z_1, 1) \cup [0, z_2)$, the count is $N - C_N(\alpha; z_1) + C_N(\alpha; z_2)$, and $N - Nz_1 + Nz_2 - Nr = 0$ because $z_2 = z_1 + r - 1$. □

When $\beta, r \in \mathbb{Q}$, Proposition 2.4 applies to each nonzero endpoint, and with the convention $\ell_k(0) := 0$ we obtain for irrational α

$$D_N(\alpha, \beta; r) = \sum_{k=0}^{J-1} (-1)^k L_k, \qquad L_k := \ell_k(z_2) - \ell_k(z_1), \qquad |L_k| \le 2(a_{k+1} + 1). \tag{2.4}$$

3. Rational endpoints: a finite extension of a Markov refinement

Throughout this section $Q \ge 2$ is fixed, and we consider one or two endpoints $z^{(i)} = P_i/Q \in (0,1)$, $P_i \in \{1, \dots, Q-1\}$, with $\gcd(P_1, P_2, Q) = 1$ (with one endpoint, $\gcd(P_1, Q) = 1$). In the situation of Theorem 1.1 these are the nonzero elements of $\{z_1, z_2\}$ and Q is their least common denominator. Write $z_k^{(i)}$ for the orbit of $z^{(i)}$ (Section 2). Constants in this section may depend on Q and P_i.

3.1. **The structure of the endpoint orbits.** Let $w_k = \big((-1)^k q_{k-1}, (-1)^k q_k\big) \bmod Q \in \mathrm{Prim}_Q$ (it is primitive since $\gcd(q_{k-1}, q_k) = 1$). Then $w_0 = (0, 1)$ and

$$w_{k+1} = w_k A_{a_{k+1}}, \qquad A_a = \begin{pmatrix} 0 & -1 \\ -1 & -a \end{pmatrix}, \tag{3.1}$$

since $w_k A_a = (-1)^k(-q_k, -q_{k-1} - aq_k)$. The matrix A_a depends only on $a \bmod Q$, and $\det A_a = -1$.

Lemma 3.1. *Let $z = P/Q \in (0,1)$ and let α be irrational. There are $v_k \in Q^{-1}\mathbb{Z}$ such that, with $u_k = \{P(w_k)_2/Q\} = \{(-1)^k P q_k/Q\}$,*

(i) *$z_k = \{u_k + v_k\alpha_k\}$ and $z_k \in (0,1)$;*
(ii) *$v_k \equiv P(w_k)_1/Q \pmod 1$;*
(iii) *$v_0 = 0$, $v_{k+1} = n_k - u_k$ with $n_k = \lfloor u_k + v_k\alpha_k \rfloor$, and $u_{k+1} = \{-u_k a_{k+1} - v_k\}$;*
(iv) *$-2 - 2\alpha_{k-1} < v_k \le 0$ for $k \ge 1$; in particular $-4 < v_k \le 0$ for all k.*

Proof. Define v_k by (iii). Then (i)–(ii) hold for $k = 0$. Assume them for k and write $z_k = u_k + v_k\alpha_k - n_k$. Since $1/\alpha_k = a_{k+1} + \alpha_{k+1}$,

$$-\frac{z_k}{\alpha_k} = -(u_k - n_k)(a_{k+1} + \alpha_{k+1}) - v_k \equiv \big(-u_k a_{k+1} - v_k\big) + (n_k - u_k)\alpha_{k+1} \pmod 1,$$

so $z_{k+1} = \{u_{k+1} + v_{k+1}\alpha_{k+1}\}$ provided $u_{k+1} \equiv -u_k a_{k+1} - v_k$; and indeed, by (3.1) and (ii), $P(w_{k+1})_2/Q = P(-(w_k)_1 - a_{k+1}(w_k)_2)/Q \equiv -v_k - a_{k+1}u_k$. Also $v_{k+1} = n_k - u_k \equiv -P(w_k)_2/Q = P(w_{k+1})_1/Q$, which is (ii). That $z_k \ne 0$ follows from $z \notin \mathbb{Z} + \alpha\mathbb{Z}$ and Lemma 2.3. For (iv): $v_{k+1} = v_k\alpha_k - z_k \in (v_k\alpha_k - 1, v_k\alpha_k]$. Hence $v_k \le 0$ for all k, $v_1 = -z > -1$, and if $v_k > -2 - 2\alpha_{k-1}$ then $v_{k+1} > -2\alpha_k - 2\alpha_{k-1}\alpha_k - 1 > -2\alpha_k - 2$ by (2.1). □

We write ζ_k for the vector of the pairs $(u_k^{(i)}, v_k^{(i)})$ attached to the endpoints $z^{(i)}$. By Lemma 3.1(ii) and the definition of $u_k^{(i)}$, $(Qv_k^{(i)}, Qu_k^{(i)}) \equiv P_i w_k \pmod{Q}$. Choosing integers e_i with $\sum_i e_i P_i \equiv 1 \pmod{Q}$ (possible since $\gcd(P_1, P_2, Q) = 1$) we get $w_k \equiv \sum_i e_i (Qv_k^{(i)}, Qu_k^{(i)})$: *the residue vector w_k is a function of ζ_k.*

3.2. **The refined partition.** Let $D_0 = 4Q$, $E_0 = \{j/d : 1 \le j < d \le D_0\}$ and $B = E_0 \cup \{1/a : a \ge 1\}$. A *piece* is a connected component of $(0,1) \setminus B$; each piece $\mathfrak{p}$ lies in a branch $(\frac{1}{a+1}, \frac{1}{a})$ of G, and we put $a(\mathfrak{p}) = a$. For $a \ge D_0$ the branch $(\frac{1}{a+1}, \frac{1}{a})$ contains no point of E_0 (a point $j/d < 1/D_0$ with $d \le D_0$ would have $j < 1$), so it is a single piece; we call these pieces *large*.

Lemma 3.2. (a) *For $u \in Q^{-1}\mathbb{Z} \cap [0,1)$ and $v \in Q^{-1}\mathbb{Z} \cap (-4, 0]$ the function $x \mapsto \lfloor u + vx \rfloor$ is constant on every piece.* (b) *G maps every piece $\mathfrak{p}$ increasingly or decreasingly onto an open interval $G(\mathfrak{p})$ whose endpoints lie in $E_0 \cup \{0,1\}$; for a large piece $G(\mathfrak{p}) = (0,1)$. The set $\mathfrak{J} = \{G(\mathfrak{p})\} \cup \{(0,1)\}$ is finite, and every $I \in \mathfrak{J}$ is, up to a countable set, the union of the pieces it contains.*

Proof. (a) The function jumps only where $u + vx \in \mathbb{Z}$, i.e. (for $v \ne 0$) at $x = (Qm - Qu)/(Qv)$ with $m \in \mathbb{Z}$; since $0 < -Qv < D_0$, such points of $(0,1)$ lie in E_0. (b) The endpoints of a piece in the branch of a are points of E_0 or the branch endpoints $\frac{1}{a+1}, \frac{1}{a}$, at which G has the one-sided limits 1 and 0. For $j/d \in E_0$ in the interior of the branch, $G(j/d) = \{d/j\}$ is a fraction with denominator $j < D_0$, hence lies in $E_0 \cup \{0\}$. The set $\mathfrak{J}$ is finite because E_0 is. An open interval with endpoints in $E_0 \cup \{0,1\}$ is the union of the pieces it contains and of the points of B in it. □

For irrational α let $\mathfrak{p}_k$ be the piece containing α_k (note $\alpha_k \notin B$), and let $\mathcal{F}'_k = \sigma(\mathfrak{p}_0, \dots, \mathfrak{p}_{k-1})$; this contains $\mathcal{F}_k = \sigma(a_1, \dots, a_k)$ because $a_{j+1} = a(\mathfrak{p}_j)$. By Lemma 3.2(a), $n_k^{(i)} = \lfloor u_k^{(i)} + v_k^{(i)} \alpha_k \rfloor$ is a function of $(\zeta_k, \mathfrak{p}_k)$, so by Lemma 3.1(iii) there are maps $f_{\mathfrak{p}}$ with

$$\zeta_{k+1} = f_{\mathfrak{p}_k}(\zeta_k); \qquad \text{hence } \zeta_k \text{ is } \mathcal{F}'_k\text{-measurable.} \tag{3.2}$$

Let $\omega_k = q_{k-1}/q_k$ ($\omega_0 = 0$), and $\mathcal{I}_k = G(\mathfrak{p}_{k-1})$ ($\mathcal{I}_0 = (0,1)$).

Lemma 3.3 (Refined cylinders)**.** *Let $\mathfrak{p}_0, \dots, \mathfrak{p}_{k-1}$ be pieces and $Z = \{\alpha : \alpha_j \in \mathfrak{p}_j \ (j < k)\} \ne \emptyset$. Then $\alpha \mapsto \alpha_k$ maps Z bijectively onto $G(\mathfrak{p}_{k-1})$ (onto $(0,1)$ if $k = 0$), with inverse $t \mapsto \frac{p_k + p_{k-1} t}{q_k + q_{k-1} t}$, where p_j, q_j are formed with the digits $a(\mathfrak{p}_0), \dots, a(\mathfrak{p}_{k-1})$. Consequently, conditionally on $\mathcal{F}'_k$, α_k has density proportional to $(1 + \omega_k t)^{-2}$ on $\mathcal{I}_k$.*

Proof. By induction on k. If G^{k-1} maps $Z' = \{\alpha_j \in \mathfrak{p}_j,\ j < k-1\}$ onto $G(\mathfrak{p}_{k-2})$, then $\mathfrak{p}_{k-1}$ meets, hence (Lemma 3.2(b)) is contained in, $G(\mathfrak{p}_{k-2})$, so G^{k-1} maps Z onto $\mathfrak{p}_{k-1}$ and G^k maps Z onto $G(\mathfrak{p}_{k-1})$. The inverse is the restriction of the inverse branch of G^k on the cylinder of the digits, which is the stated Möbius map; its derivative is $\pm(q_k + q_{k-1} t)^{-2}$. □

3.3. **The extended chain.** For $\omega \in [0,1]$ and $I \in \mathfrak{J}$ let $K_{\omega,I}$ be the law of the piece containing a point $t \in I$ drawn with density proportional to $(1 + \omega t)^{-2}$. Since $(1 + \omega t)^{-2} \in [\frac{1}{4}, 1]$,

$$K_{\omega,I}(\mathfrak{p}) \ge \frac{|\mathfrak{p}|}{4|I|} \ge \frac{|\mathfrak{p}|}{4} \qquad (\mathfrak{p} \subset I), \qquad K_{\omega,(0,1)}\big((\tfrac{1}{a+1}, \tfrac{1}{a})\big) \ge \frac{1}{4a(a+1)}. \tag{3.3}$$

Let $W_k = (\omega_k, \mathcal{I}_k, \zeta_k)$. By Lemma 3.3, given $\mathcal{F}'_k$ the piece $\mathfrak{p}_k$ has law $K_{\omega_k, \mathcal{I}_k}$, and then $W_{k+1} = (1/(a(\mathfrak{p}_k) + \omega_k), G(\mathfrak{p}_k), f_{\mathfrak{p}_k}(\zeta_k))$. Hence (W_k) is a time-homogeneous Markov chain with respect to $(\mathcal{F}'_k)$, with kernel

$$\mathsf{K}\big((\omega, I, \zeta), \cdot\big) = \text{law of } \big(1/(a(\mathfrak{p}) + \omega),\ G(\mathfrak{p}),\ f_{\mathfrak{p}}(\zeta)\big), \qquad \mathfrak{p} \sim K_{\omega,I},$$

and the conditional law of $(\mathfrak{p}_k, \mathfrak{p}_{k+1}, \dots)$ given $\mathcal{F}'_k$ is the law of the pieces emitted by this chain started from W_k. Let $\mathcal{W}$ be the (finite) set of pairs $(\mathcal{I}_k(\alpha), \zeta_k(\alpha))$, $\alpha \in (0,1) \setminus \mathbb{Q}$, $k \ge 0$. If

$(I,\zeta)=(\mathcal{I}_k(\alpha),\zeta_k(\alpha))$ and $\mathfrak{p}\subset I$, then by Lemma 3.3 some α' with the same first k pieces has $\alpha'_k\in\mathfrak{p}$, so $(G(\mathfrak{p}),f_{\mathfrak{p}}(\zeta))\in\mathcal{W}$. Thus the chain started from $[0,1]\times\mathcal{W}$ stays there, and we always consider such starting points.

Lemma 3.4 (Generation)**.** *There is an even n_Q such that every $g\in\mathrm{SL}_2(\mathbb{Z}/Q\mathbb{Z})$ is a product $A_{c_1}\cdots A_{c_{n_Q}}$ with $c_j\in\mathbb{Z}/Q\mathbb{Z}$. Moreover $\mathrm{SL}_2(\mathbb{Z}/Q\mathbb{Z})$ acts transitively on Prim_Q by right multiplication.*

Proof. $A_0^2=I$, $A_0A_c=\left(\begin{smallmatrix}1&c\\0&1\end{smallmatrix}\right)$ and $A_cA_0=\left(\begin{smallmatrix}1&0\\c&1\end{smallmatrix}\right)$. The reduction map $\mathrm{SL}_2(\mathbb{Z})\to\mathrm{SL}_2(\mathbb{Z}/Q\mathbb{Z})$ is surjective and $\mathrm{SL}_2(\mathbb{Z})$ is generated by the two elementary matrices with $c=1$, whose inverses are the elementary matrices with $c=-1$; so every element of the finite group $\mathrm{SL}_2(\mathbb{Z}/Q\mathbb{Z})$ is a product of an even number $\le n_Q$ of the A_c, and we pad with A_0A_0. For transitivity it suffices to show that every $(b,c)\in\mathrm{Prim}_Q$ is congruent to a primitive vector of $\mathbb{Z}^2$ (which is the second row of a matrix in $\mathrm{SL}_2(\mathbb{Z})$, and $(0,1)$ times that matrix is the vector). Represent c by an integer $c\neq0$ and let t be the product of the primes dividing c but not b. If a prime $p\mid c$ divides b, then $p\nmid Q$ and $p\nmid t$, so $p\nmid b+tQ$; otherwise $p\mid t$ and $b+tQ\equiv b\not\equiv0\pmod p$. Thus $\gcd(b+tQ,c)=1$. □

Lemma 3.5 (Coupling)**.** *There are constants $C_Q<\infty$ and $\vartheta_Q\in(0,1)$ with the following property. For any two states $W=(\omega,I,\zeta)$ and $W'=(\omega',I',\zeta')$ with $(I,\zeta),(I',\zeta')\in\mathcal{W}$ there is a coupling of the chains started at W and W', and a random time τ with $\mathbb{P}(\tau>n)\le C_Q\vartheta_Q^n$, such that for all $k\ge\tau$ the emitted pieces coincide, $(\mathcal{I}_k,\zeta_k)=(\mathcal{I}'_k,\zeta'_k)$, and $|\omega_k-\omega'_k|\le F_{k-\tau+1}^{-2}$, where F_j are the Fibonacci numbers.*

Proof. Step 1 (reaching the full interval). Every $I\in\mathfrak{J}$ contains a cylinder $\{a_1=b_1,\dots,a_m=b_m\}$, and the points of this cylinder with $a_{m+1}=D_0$ form a non-empty open interval; the pieces of $t,Gt,\dots,G^mt$ for an irrational t in it form an admissible sequence $\mathfrak{p}^I_1,\dots,\mathfrak{p}^I_{m(I)}$ ending with a large piece. By (3.3) a chain at any state with interval I emits this sequence with probability at least $\prod_j|\mathfrak{p}^I_j|/4>0$; afterwards its interval is $(0,1)$, and by (3.3) it can emit the large piece of digit D_0 with probability $\ge(4D_0(D_0+1))^{-1}$ at each further step. Hence there are m_1 and $c_1>0$ such that from any state, with probability $\ge c_1$, after exactly m_1 steps the chain has interval $(0,1)$.

Step 2 (synchronising ζ). Consider a chain with interval $(0,1)$ which emits n_Q+1 large pieces with digits $b_j\in[D_0,D_0+Q)$. For a large digit b and $x=\alpha_k\in(\frac{1}{b+1},\frac1b)$ we have $|v^{(i)}x|<4/D_0=1/Q$, so if $u^{(i)}\neq0$ then $u^{(i)}\in[\frac1Q,1-\frac1Q]$ gives $\lfloor u^{(i)}+v^{(i)}x\rfloor=0$ and the new $v^{(i)}$ equals $-u^{(i)}$. By Lemma 3.4 and (3.1), given the current w we can choose $b_1,\dots,b_{n_Q}$ (as residues modulo Q) such that w is mapped to $(0,1)$; then $u^{(i)}=\{P_i/Q\}\neq0$ for every endpoint, and after one more step with $b_{n_Q+1}=D_0$ the new state is $w=(0,1)A_{D_0}=(-1,0)$, $u^{(i)}=0$, $v^{(i)}=-P_i/Q$, independently of the state before the block. Each such sequence of n_Q+1 large pieces is emitted with probability $\ge c_2:=(4(D_0+Q)^2)^{-n_Q-1}$.

Step 3 (staying together). Suppose that at some time both chains have the same (I,ζ). At each subsequent step draw the two pieces from a maximal coupling of $K_{\omega,I}$ and $K_{\omega',I}$. When they agree, (I,ζ) stays equal and the common digit b maps ω,ω' to $1/(b+\omega)$, $1/(b+\omega')$. The densities $(1+\omega t)^{-2}/Z_\omega$ satisfy $|\partial_\omega\log((1+\omega t)^{-2}/Z_\omega)|\le4$, so the probability of disagreement is at most $\min(\frac34,8|\omega-\omega'|)$ (the bound $\frac34$ follows from (3.3)). After j agreements with digits $b_1,\dots,b_j$, ω and ω' are the values at ω_0,ω'_0 of $s\mapsto[0;b_j,\dots,b_2,b_1+s]$, whose derivative is at most the inverse square of the continuant of $b_1,\dots,b_j$, which is $\ge F_{j+1}$. Hence all agreements succeed with probability at least $c_3=\prod_{j\ge0}\max(\frac14,1-8F_{j+1}^{-2})>0$.

Renewal. An *attempt* consists of m_1+n_Q+1 steps in which the two chains move independently, followed, if both chains have followed Steps 1 and 2, by Step 3. A new attempt is started whenever the current one fails (at the end of the independent phase, or at the first disagreement in Step 3). This is a Markovian coupling with the correct marginals. Each attempt succeeds with conditional

probability $\geq \delta = (c_1c_2)^2c_3$ given the past, and a failed attempt lasts $m_1 + n_Q + 1 + j$ steps, where j is the step of Step 3 at which it fails, with $\mathbb{P}(j \geq j_0) \leq \sum_{j\geq j_0} 8F_{j+1}^{-2}$. For $\eta > 0$ small enough the conditional expectation of $e^{\eta\cdot\text{duration}}$ over a failed attempt is at most $1-\delta/2$, so the starting time τ of the successful Step 3 satisfies $\mathbb{E}e^{\eta\tau} < \infty$, uniformly in the initial states. From τ on everything agrees, and $|\omega_k - \omega'_k| \leq F_{k-\tau+1}^{-2}$ by Step 3. □

3.4. **Consequences.** By a *functional of the chain from time m* we mean a random variable $\Phi = \phi(\omega_m, \mathcal{I}_m, \zeta_m, \mathfrak{p}_m, \mathfrak{p}_{m+1}, \dots)$ with ϕ Borel; it is called Λ-*regular* if $|\phi| \leq 1$ and ϕ is Λ-Lipschitz in its first argument. $\Phi^{[m']}$ denotes the same ϕ evaluated at time m'.

Lemma 3.6 (Mixing)**.** *There are constants C'_Q and $\vartheta'_Q \in (0,1)$ such that for every Λ-regular functional Φ of the chain from time m:*

(a) $\big|\mathbb{E}[\Phi \mid \mathcal{F}'_j] - \mathbb{E}\Phi\big| \leq C'_Q(1+\Lambda){\vartheta'_Q}^{m-j}$ *almost surely, for $0 \leq j \leq m$;*

(b) *the limit $\tilde{\mu}(\phi) = \lim_{m'\to\infty} \mathbb{E}\Phi^{[m']}$ exists and $|\mathbb{E}\Phi^{[m']} - \tilde{\mu}(\phi)| \leq C'_Q(1+\Lambda){\vartheta'_Q}^{m'}$.*

Proof. (a) By the Markov property $\mathbb{E}[\Phi \mid \mathcal{F}'_j] = g(W_j)$ with $g(W) = \mathbb{E}_W\Phi^{[m-j]}$, the chain being started at W at time 0. For two states coupled as in Lemma 3.5, $|g(W) - g(W')| \leq 2\mathbb{P}(\tau > n) + \Lambda\,\mathbb{E}\big[F_{n-\tau+1}^{-2}; \tau \leq n\big]$ with $n = m - j$, which is $\leq C(1+\Lambda)\vartheta^n$ for some $\vartheta < 1$ (split according to $\tau \leq n/2$). As $\mathbb{E}\Phi = \mathbb{E}g(W_j)$, the claim follows. (b) $\mathbb{E}\Phi^{[m'+1]} = \mathbb{E}h(W_1)$ and $\mathbb{E}\Phi^{[m']} = h(W_0)$ with $h(W) = \mathbb{E}_W\Phi^{[m']}$; the same coupling bound shows that $(\mathbb{E}\Phi^{[m']})$ is a Cauchy sequence with the stated rate. □

Since the kernel does not depend on the parity of the time, *even and odd times have the same limit $\tilde{\mu}$*; this is the source of the cancellation of drifts in Sections 8 and 9.

Lemma 3.7 (The stationary law of (ω, w))**.** *Let $\gamma(dt) = \frac{dt}{(1+t)\log 2}$ be the Gauss measure and U_Q the uniform law on Prim_Q. For $f : [0,1] \times \mathrm{Prim}_Q \to [-1,1]$ which is Λ-Lipschitz in the first variable, $\big|\mathbb{E}f(\omega_k, w_k) - \int f\, d(\gamma \otimes \mathrm{U}_Q)\big| \leq C'_Q(1+\Lambda){\vartheta'_Q}^k$. In particular $c_k := (w_k)_2$ converges in law to c_Q, and $\tilde{\mu}\big(g(\omega)h(c)\big) = \int g\, d\gamma \cdot \mathbb{E}h(c_Q)$.*

Proof. If t has density $(1+\omega)(1+\omega t)^{-2}$ on $(0,1)$ and $\omega \sim \gamma$, the pair (t,ω) has density $(\log 2)^{-1}(1+t\omega)^{-2}$, which is invariant under $(t,\omega) \mapsto (Gt, 1/(a_1(t)+\omega))$ (the natural extension of G; with $t = 1/(a+t')$, $\omega = 1/\omega' - a$ one has $1 + t\omega = (1+t'\omega')/(\omega'(a+t'))$, and the Jacobian is $(a+t')^{-2}\omega'^{-2}$); its second marginal is γ. Hence the digit-driven chain $\omega \mapsto 1/(a+\omega)$, $w \mapsto wA_a$, $a \sim$ law of $a_1(t)$, preserves $\gamma \otimes \mathrm{U}_Q$ ($w \mapsto wA_a$ is a bijection of Prim_Q). For each $w \in \mathrm{Prim}_Q$ choose $\zeta(w)$ with $((0,1), \zeta(w)) \in \mathcal{W}$ and residue vector w (steer with large digits as in Step 2). Started from $(\omega, (0,1), \zeta(w))$ with $(\omega, w) \sim \gamma \otimes \mathrm{U}_Q$, the extended chain emits the pieces of the orbit of a point t with density $(1+\omega)(1+\omega t)^{-2}$: this is the computation of Lemma 3.3 with the prior density of t included, which replaces q_j by $q_j + \omega p_j$ and hence ω_k by $(q_{k-1} + \omega p_{k-1})/(q_k + \omega p_k) = [0; a_k, \dots, a_2, a_1 + \omega]$, the value produced by the kernel. So its (ω_k, w_k) is the stationary digit-driven chain. Coupling it with the actual chain (Lemma 3.5) gives the estimate. □

Lemma 3.8. *For $Q \geq 2$, $\sum_{(b,c)\in\mathrm{Prim}_Q} B_2\big(\{c/Q\}\big) = 0$, where $B_2(x) = x^2 - x + \frac{1}{6}$. Consequently $\mathbb{E}\big[\{Pc_Q/Q\}(1 - \{Pc_Q/Q\})\big] = \frac{1}{6}$ for every P coprime to Q.*

Proof. By Möbius inversion over $d \mid \gcd(b,c,Q)$ and the distribution relation $\sum_{j=0}^{m-1} B_2(j/m) = \frac{1}{6m}$,

$$\sum_{(b,c)\in\mathrm{Prim}_Q} B_2(\{c/Q\}) = \sum_{d\mid Q} \mu(d)\,\frac{Q}{d} \sum_{j \bmod Q/d} B_2\Big(\frac{j}{Q/d}\Big) = \frac{1}{6}\sum_{d\mid Q}\mu(d) = 0.$$

Since $\mathbb{P}(c_Q = c)$ depends only on $\gcd(c,Q)$, Pc_Q has the same law as c_Q; and $x(1-x) = \frac{1}{6} - B_2(x)$. □

4. Cells

Fix N. For $j \geq 0$ and irrational α let the *level-j cell* of α be

$$\mathcal{C}_j(\alpha) = \{\alpha' \in (0,1) \setminus \mathbb{Q} : \ a_l(\alpha') = a_l(\alpha),\ N_l(\alpha') = N_l(\alpha)\ (1 \leq l \leq j)\},$$

so $\mathcal{C}_0 = (0,1) \setminus \mathbb{Q}$. Let $\mathcal{H}_j$ be the σ-algebra generated by the level-j cells. Clearly $\mathcal{H}_j \subset \mathcal{H}_{j+1}$, and J is an $(\mathcal{H}_j)$-stopping time, since $\{J = j\} = \{N_{j-1} \geq 1, N_j = 0\} \in \mathcal{H}_j$.

Lemma 4.1. *Every cell is the set of irrational points of an open interval. On a level-j cell C ($j \geq 1$) with $N_{j-1} \geq 1$ the maps $\alpha \mapsto \alpha_{j-1}$ and $\alpha \mapsto Y_{j-1} = N_{j-1}\alpha_{j-1}$ are continuous and strictly monotone, $Y_{j-1}(C) \subset [N_j, N_j + 1)$, and hence $y_{j-1} = Y_{j-1} - N_j$ maps C monotonically onto an interval $R_{j-1}(C) \subset [0,1)$, the* range *of C.*

Proof. On a cylinder $\{a_1, \ldots, a_{j-1}$ fixed$\}$ the map $\alpha \mapsto \alpha_{j-1}$ is a Möbius map, $\alpha = \frac{p_{j-1} + p_{j-2}\alpha_{j-1}}{q_{j-1} + q_{j-2}\alpha_{j-1}}$. By induction on j: inside a level-$(j-1)$ cell (an interval on which N_{j-1} is constant), $a_j = \lfloor 1/\alpha_{j-1} \rfloor$ and $N_j = \lfloor N_{j-1}\alpha_{j-1} \rfloor$ are monotone step functions of the monotone function α_{j-1}, so their level sets are intervals. □

Lemma 4.2 (Fibre relation)**.** *Let C be a level-j cell with $N_j \geq 1$, and put $M = N_{j-1} - a_j N_j \in \mathbb{Z}$, $\Phi(u) = \frac{N_{j-1}u}{N_j + u}$. On C,*

$$\alpha_{j-1} = \frac{N_j + y_{j-1}}{N_{j-1}}, \qquad Y_j = M - \Phi(y_{j-1}), \qquad \frac{1}{\Phi'(y_{j-1})} = \alpha_{j-1}\Big(1 + \frac{y_{j-1}}{N_j}\Big).$$

Proof. $Y_{j-1} = N_j + y_{j-1} = N_{j-1}\alpha_{j-1}$, and $Y_j = N_j(1/\alpha_{j-1} - a_j) = \frac{N_j N_{j-1}}{N_j + y_{j-1}} - a_j N_j = M - \Phi(y_{j-1})$. Finally $\Phi'(u) = \frac{N_{j-1}N_j}{(N_j+u)^2}$ and $N_j + y_{j-1} = N_{j-1}\alpha_{j-1}$. □

The endpoints of cells are rational. An endpoint e of a level-j cell is a point at which some a_l or some N_l ($l \leq j$) is discontinuous. We call a point e an *integer point of level i* if $a_1, \ldots, a_i$ and $N_1, \ldots, N_{i-1}$ are constant on a neighbourhood of e and $\lim_{\alpha \to e} Y_{i-1}(\alpha) \in \mathbb{Z}$ (the depth to which the digits are constant near e is an additional hypothesis in the lemma below). For a function F continuous near e we write $F(e)$ for its limit at e.

Lemma 4.3 (Integer points)**.** *Let e be an integer point of level i, $\lim_{\alpha \to e} Y_{i-1} = n + 1$, and suppose $a_1, \ldots, a_{m+1}$ are constant near e, where $m \geq i$, and that $N_l \geq 2$ for all $l \leq m$ on one side of e (near e). Call A the side of e on which $Y_{i-1} < n + 1$ and B the other side. Then the one-sided limits at e are*

	y_{i-1}	y_i	y_l $(i < l \leq m)$
A	1	$1 - \alpha_i(e)$	$1 - \alpha_l(e)$
B	0	1	$1 - \alpha_l(e)$

and N_l is the same on both sides for $l \neq i$, $l \leq m + 1$ (while $N_i = n$ on A, $n + 1$ on B).

Proof. At e, $\alpha_{i-1}(e) = \frac{n+1}{N_{i-1}}$, so $c := (n+1)\alpha_i(e) = N_{i-1} - a_i(n+1) \in \mathbb{Z}$. On A, $N_i = n$, $y_{i-1} \to 1$, and $Y_i = n\alpha_i \to c - \alpha_i(e)$, so $N_{i+1} = c - 1$ and $y_i \to 1 - \alpha_i(e)$. On B, $N_i = n + 1$, $y_{i-1} \to 0$; as α_{i-1} decreases towards $\alpha_{i-1}(e)$ from above, α_i increases to $\alpha_i(e)$, so $Y_i = (n+1)\alpha_i \uparrow c$, whence $N_{i+1} = c - 1$ and $y_i \to 1$. In both cases $-y_i/\alpha_i$ tends to a non-integer ($1 - a_{i+1} - \alpha_{i+1}(e)$, resp. $-a_{i+1} - \alpha_{i+1}(e)$), so by Proposition 2.4, $y_{i+1} \to \{-1/\alpha_i(e)\} = 1 - \alpha_{i+1}(e)$ on both sides; inductively $y_l \to 1 - \alpha_l(e)$ for $i < l \leq m$, these limits being non-integers. Since N_{i+1} agrees on both sides and $Y_l = N_l\alpha_l$ is continuous with non-integer limit for $l > i$, the N_l agree for $l > i$. In particular, as $N_i \in \{n, n+1\}$ and the other N_l agree, $N_l \geq 1$ for $l \leq m$ on both sides, so no absorption occurs and the computation above is legitimate on both sides. □

Definition 4.4. Let $1 \leq i \leq k + 1$. A level-i cell C is *k-regular* if $a_1, \ldots, a_{k+2}$ are constant on an open interval containing the closure of C, and $N_l \geq 2$ on C for all $l \leq k + 1$.

Let C be a k-regular level-i cell and e an endpoint of a level-l sub-cell $C' \subset C$, $i \le l \le k+1$. Since $e \in \overline{C}$, the digits $a_1, \dots, a_{k+2}$ are constant near e, so only some N_j ($j \le l$) is discontinuous at e. If i' is the least such j, then $N_1, \dots, N_{i'-1}$ are constant near e, $Y_{i'-1} = N_{i'-1}\alpha_{i'-1}$ is continuous near e, and $N_{i'} = \lfloor Y_{i'-1} \rfloor$ jumps, so $\lim_{\alpha \to e} Y_{i'-1} \in \mathbb{Z}$: e is an integer point of level $i' \le l$, and since $N_j \ge 2$ ($j \le k+1$) on the side of e inside C', Lemma 4.3 applies at e with $m = k+1$. Every level-l sub-cell ($i \le l \le k+1$) of a k-regular level-i cell is again k-regular.

Proposition 4.5 (Three types)**.** *Let C be a level-j cell whose endpoints are integer points (of levels $\le j$) satisfying the hypotheses of Lemma 4.3 with $m \ge j$, and assume $N_{j-1} \ge 2$ on C. Then, up to its endpoints, $R_{j-1}(C)$ is one of*

$$\boldsymbol{F}\colon\ [0,1), \qquad \boldsymbol{P}\colon\ (1-\alpha_{j-1}(e),1), \qquad \boldsymbol{Q}\colon\ [0,1-\alpha_{j-1}(e)),$$

where e is an endpoint of C. In particular this holds for every level-j sub-cell of a k-regular cell, $j \le k+1$.

Proof. $R_{j-1}(C)$ is the interval between the one-sided limits of y_{j-1} at the two endpoints e_1, e_2 of C. By Lemma 4.3 (applied with the level $i \le j$ of the endpoint, and $l = j-1 \ge i-1$) each limit belongs to $\{0, 1, 1-\alpha_{j-1}(e_\nu)\}$. Since $Y_{j-1} = N_{j-1}\alpha_{j-1}$ on C, $|R_{j-1}(C)| = N_{j-1}|\alpha_{j-1}(e_1) - \alpha_{j-1}(e_2)| > 0$. If both limits were of the third kind we would get $|R_{j-1}(C)| = |R_{j-1}(C)|/N_{j-1}$, impossible for $N_{j-1} \ge 2$. If both lie in $\{0,1\}$ the range is $[0,1)$. Otherwise it is of type $\mathbf{P}$ or $\mathbf{Q}$. □

Lemma 4.6 (Distortion)**.** *Let C be a level-j cell, $j \ge 1$, with $N_{j-1} \ge 1$. The conditional law of y_{j-1} given $\alpha \in C$ has a density on $R_{j-1}(C)$ whose supremum and infimum differ by a factor at most $(1+1/N_{j-1})^2 \le 1 + 3/N_{j-1}$.*

Proof. With $t = \alpha_{j-1} = (N_j + y_{j-1})/N_{j-1}$ we have $|d\alpha/dy_{j-1}| = N_{j-1}^{-1}(q_{j-1} + q_{j-2}t)^{-2}$, and t varies in an interval of length $\le 1/N_{j-1}$ on C; since $q_{j-2} \le q_{j-1}$ the ratio is at most $(1+1/N_{j-1})^2$. □

Lemma 4.7 (Oscillations)**.** (a) *Let $0 \le k \le i-2$ and let C be a level-i cell with $i \le J$ on C. Then*

$$\operatorname{osc}_C y_k \le e^4 \sup_C(\alpha_k\alpha_{k+1}\cdots\alpha_{i-2}) \le e^4 \min\left(2^{-\lfloor (i-1-k)/2 \rfloor},\ a_{k+1}^{-1} 2^{-\lfloor (i-2-k)/2 \rfloor}\right).$$

If $i > J$ and $k \le J-2$, $\operatorname{osc}_C y_k$ is bounded by the same expression with i replaced by J. (b) *Let C be a level-i cell, $i \ge 1$, with $N_{i-1} \ge 1$, on which $a_i, \dots, a_{l+1}$ are constant ($l \ge i-1$). Then $\operatorname{osc}_C \alpha_l \le 4\,Q^2/N_{i-1}$, where $Q = \prod_{t=i}^{l}(a_t+1)$ (empty product $= 1$).*

Proof. (a) On C all $N_1, \dots, N_i$ are fixed, y_{i-1} ranges in an interval of length ≤ 1, and by Lemma 4.2 (applied at levels $l = k+1, \dots, i-1$) y_k is a smooth monotone function of y_{i-1} with $|dy_k/dy_{i-1}| = \prod_{l=k}^{i-2} \alpha_l(1 + y_l/N_{l+1})$. Since $N_{l+2} \le N_l/2$, $\sum_{l>k} N_l^{-1} \le 4/N_{i-1} \le 4$. The product of r consecutive α's is at most $2^{-\lfloor r/2 \rfloor}$ and $\alpha_k < 1/a_{k+1}$. The case $i > J$ follows because cells decrease with the level. (b) On C, α_{i-1} varies in an interval of length $\le 1/N_{i-1}$ (Lemma 4.2); writing $\alpha_{i-1} = \frac{P + P'\alpha_l}{Q_0 + Q_0'\alpha_l}$ with the convergent denominators $Q_0' \le Q_0 \le Q$ of $[0; a_i, \dots, a_l]$, we get $|d\alpha_l/d\alpha_{i-1}| = (Q_0 + Q_0'\alpha_l)^2 \le 4Q^2$. □

Remark 4.8. By Lemma 4.2, for regular cells the children of a cell of type $\mathbf{F}$, $\mathbf{P}$, $\mathbf{Q}$ at level j have types whose conditional probabilities are, up to relative errors of order $(a_j+1)(a_{j+1}+1)/N_j$, those of a three-state Markov chain with transition probabilities (with $x = \alpha_{j-1}$, $a = a_j$, $y = \alpha_j$) $\mathbf{F} \to \mathbf{F}$: ax, $\mathbf{F} \to \mathbf{P}$: xy; $\mathbf{P} \to \mathbf{P}$: y, $\mathbf{P} \to \mathbf{Q}$: $1-y$; $\mathbf{Q} \to \mathbf{F}$: $\frac{(a-1)x}{1-x}$, $\mathbf{Q} \to \mathbf{P}$: $\frac{xy}{1-x}$. Its three-step Dobrushin coefficient can be arbitrarily close to 1 (digits $(1, A, 1, \dots)$ with A large), although numerically the coefficients over $n \ge 4$ steps appear to be bounded by $1/F_n$. The proof below does not use this chain: it uses only the induced law of y, which is described by the scalar recursion of Section 5 with an explicit two-step contraction.

5. The two-piece transfer

For $x \in (0,1)$ and $p \in [0,1]$ let $\nu_{x,p}$ be the probability measure on $[0,1)$ with density $\frac{1-p}{1-x}$ on $[0,1-x)$ and $\frac{p}{x}$ on $[1-x,1)$ (the *x-two-piece law with upper mass p*); $\nu_{x,x}$ is Lebesgue measure. For x with $1/x \notin \mathbb{Z}$ write $a = \lfloor 1/x \rfloor$, $y = \{1/x\}$, $T_x(u) = \{-u/x\}$, and

$$\lambda(x) = -\frac{y(1-y)}{a+y-1}, \qquad \mu(x) = \frac{ay}{a+y-1}. \tag{5.1}$$

Lemma 5.1 (Exact transfer)**.** $T_{x*}\,\nu_{x,p} = \nu_{y,\ \lambda(x)p+\mu(x)}$.

Proof. Let $A = \frac{1-p}{1-x}$, $B = \frac{p}{x}$. The density of the image is $g(v) = x\sum_c f(x(c-v))$, the sum over the integers c with $x(c-v) \in [0,1)$. Note $x(a-1) \le 1-x \le xa$. For $v \in (0,1-y)$ the admissible c are $1,\dots,a$; for $c \le a-1$, $x(c-v) < 1-x$, while for $c = a$, $x(a-v) > x(a-1+y) = 1-x$. For $v \in (1-y,1)$ the admissible c are $1,\dots,a+1$; now $x(a-v) < 1-x$ and $x(a+1-v) \in (xa,1) \subset [1-x,1)$. Hence $g = x((a-1)A+B)$ on $(0,1-y)$ and $g = x(aA+B)$ on $(1-y,1)$, so the image is y-two-piece with upper mass $yx(aA+B) = y\big(\frac{a(1-p)}{a+y-1} + p\big) = \mu(x) + \lambda(x)p$, using $x = \frac{1}{a+y}$. ☐

Let $(x_l)_{l \ge l_0}$ be a Gauss orbit, $x_{l+1} = G(x_l)$, with digits $b_{l+1} = \lfloor 1/x_l \rfloor$, and write $\lambda_{l+1} = \lambda(x_l)$, $\mu_{l+1} = \mu(x_l)$ (these depend on b_{l+1} and x_{l+1}). For $p_{l_0} \in [0,1]$ let

$$p_{l+1} = \lambda_{l+1}p_l + \mu_{l+1} \qquad (l \ge l_0). \tag{5.2}$$

Lemma 5.2 (Uniform contraction)**.** *For all $l \ge l_0$:*

(i) $p_{l+1} \in [0,1]$ *and* $1-p_{l+1} \le 1-x_{l+1}$*;*
(ii) $p_l \le 2x_l$ *for* $l \ge l_0+2$*;*
(iii) $|\lambda_{l+1}| \le 1-x_{l+1}$, $|\lambda_{l+1}\lambda_{l+2}| \le 3-2\sqrt{2} < \frac{1}{2}$ *and* $|\lambda_{l+1}\lambda_{l+2}| \le x_{l+2}$*;*
(iv) *if (p'_l) is another solution of (5.2) with the same (x_l), then for $m > l \ge l_0$*

$$|p_m - p'_m| \le 2^{-\lfloor (m-l)/2 \rfloor}|p_l - p'_l|, \qquad \text{and} \qquad |p_m - p'_m| \le x_m\, 2^{-\lfloor (m-l-2)/2 \rfloor}|p_l - p'_l| \quad (m \ge l+2).$$

Proof. Write $y = x_{l+1}$, $b = b_{l+1}$, $\kappa = \frac{b}{b+y-1} \ge 1$, so $p_{l+1} = y\kappa(1-p_l) + yp_l$ and $1-p_{l+1} = (1-p_l)(1-y\kappa) + p_l(1-y)$ with $1-y\kappa = \frac{(b-1)(1-y)}{b+y-1} \in [0,1-y]$; this gives (i). For (ii): if $b_l \ge 2$ then $\kappa \le 2$ and $p_l \le x_l \max(\kappa,1) \le 2x_l$. If $b_l = 1$ then $p_l = (1-p_{l-1}) + x_l p_{l-1} \le (1-x_{l-1}) + x_l$ by (i), and $1-x_{l-1} = \frac{x_l}{1+x_l} \le x_l$. For (iii): $|\lambda_{l+1}| = \frac{y(1-y)}{b+y-1} \le 1-y$ because $b \ge 1$. Write $z = x_{l+2}$. If $b_{l+2} \ge 2$ then $|\lambda_{l+2}| \le \frac{z(1-z)}{1+z}$ and $|\lambda_{l+1}| \le 1$; if $b_{l+2} = 1$ then $|\lambda_{l+1}| \le 1-x_{l+1} = \frac{z}{1+z}$ and $|\lambda_{l+2}| \le 1-z$. In both cases $|\lambda_{l+1}\lambda_{l+2}| \le \frac{z(1-z)}{1+z} \le \min(z,\, 3-2\sqrt{2})$, the maximum of $\frac{z(1-z)}{1+z}$ on $(0,1)$ being attained at $z = \sqrt{2}-1$. (iv) follows from $p_m - p'_m = \big(\prod_{t=l+1}^{m}\lambda_t\big)(p_l - p'_l)$, grouping the factors in pairs (the last pair by the third inequality of (iii)). ☐

For later use we record two elementary properties of $\nu_{x,p}$: its total variation distance to Lebesgue measure is $|p-x|$, and its density is bounded when p is comparable to x:

$$\|\nu_{x,p} - \nu_{x,x}\|_{TV} = |p-x|, \qquad \frac{d\nu_{x,p}}{du} \le \max\Big(\frac{1-p}{1-x}, \frac{p}{x}\Big) \le 2 \quad \text{if } p \le 2x \text{ and } 1-p \le 1-x. \tag{5.3}$$

Moreover $\frac{\partial}{\partial p}\int g\, d\nu_{x,p} = \frac{1}{x}\int_{1-x}^{1} g - \frac{1}{1-x}\int_0^{1-x} g$, so $\big|\frac{\partial}{\partial p}\int g\, d\nu_{x,p}\big| \le 2\sup|g|$.

The conditional law of the fibre variables. For a level-i cell C which is k-regular ($i \le k$) fix a reference point $\alpha^* \in C$, put $x_l = \alpha_l(\alpha^*)$ and $b_l = a_l$ (constant on C for $l \le k+2$), and let p_l^C ($i-1 \le l \le k$) be the solution of (5.2) with

$$p_{i-1}^C = x_{i-1},\ 1,\ 0 \qquad \text{according as } C \text{ has type } \mathbf{F},\ \mathbf{P},\ \mathbf{Q}.$$

Define the error parameter

$$E_C = \frac{1}{\min_C N_{k+1}} + \frac{1}{N_{i-1}} \prod_{u=i}^{k+2} (b_u + 1)^5. \tag{5.4}$$

Theorem 5.3 (Conditional law of y_l)**.** *There is an absolute constant C_0 such that, for every k-regular level-i cell C and every $i-1 \le l \le k$,*

$$\left\| \mathrm{Law}(y_l \mid C) - \nu_{x_l, p_l^C} \right\|_{TV} \le C_0 \, (l - i + 2) \, E_C.$$

Proof. Write $o_l = \mathrm{osc}_C \, \alpha_l$; by Lemma 4.7(b), $o_l \le 4 \prod_{u=i}^{l} (b_u+1)^2 / N_{i-1}$. Note $1/x_l \le b_{l+1} + 1$ and $1/(1-x_l) \le 2(b_{l+2}+1)$. We may assume $C_0 E_C \le 1$, so that $o_l \le x_l/2$ and $o_l \le (1-x_l)/2$ for $l \le k$; the cell C has finitely many sub-cells, and we write $\underline{N}_l = \min_C N_l \ge \min_C N_{k+1}$ for $l \le k+1$.

Step 1. Let $i-1 \le l \le k$. The level-$(l+1)$ sub-cells C' of C partition C; by Lemma 4.6 and Proposition 4.5, $\mathrm{Law}(y_l \mid C')$ is within total variation $3/N_l$ of the uniform law on an interval of type **F**, $(1-\alpha_l(e'), 1)$ or $[0, 1-\alpha_l(e'))$ with $|\alpha_l(e') - x_l| \le o_l$. Replacing $\alpha_l(e')$ by x_l costs at most $o_l/x_l + o_l/(1-x_l)$ in total variation. The resulting mixture is x_l-two-piece. Hence $\|\mathrm{Law}(y_l \mid C) - \nu_{x_l, \tilde{p}_l}\|_{TV} \le e_l := 3/\underline{N}_l + o_l(x_l^{-1} + (1-x_l)^{-1})$ for some $\tilde{p}_l$, and $\pi_l := \mathbb{P}(y_l \ge 1 - x_l \mid C)$ satisfies $|\pi_l - \tilde{p}_l| \le e_l$. For $l = i-1$ the partition is trivial and the type of C gives $|\pi_{i-1} - p_{i-1}^C| \le 2 e_{i-1}$.

Step 2. Let $i-1 \le l < k$. On C, $y_{l+1} = \{-y_l/\alpha_l\}$ and $|y_l/\alpha_l - y_l/x_l| \le \eta_l := 2o_l/x_l^2$, so on the circle $\mathbb{R}/\mathbb{Z}$ the points y_{l+1} and $T_{x_l}(y_l)$ are within η_l. Thus $\{y_{l+1} \ge 1 - x_{l+1}\}$ and $\{T_{x_l}(y_l) \ge 1 - x_{l+1}\}$ differ only if $T_{x_l}(y_l)$ is within η_l of 0 or of $1 - x_{l+1}$. By Lemma 5.1, under $\nu_{x_l, \tilde{p}_l}$ the variable $T_{x_l}(u)$ has law $\nu_{x_{l+1}, \lambda_{l+1}\tilde{p}_l + \mu_{l+1}}$, whose density is at most $1/\min(x_{l+1}, 1-x_{l+1})$. Therefore

$$\pi_{l+1} = \lambda_{l+1}\pi_l + \mu_{l+1} + r_l, \qquad |r_l| \le 3e_l + \frac{4\eta_l}{\min(x_{l+1}, 1-x_{l+1})}.$$

Step 3. Since $|\lambda_t| \le 1$, $|\pi_l - p_l^C| \le 2e_{i-1} + \sum_{t=i-1}^{l-1} |r_t|$, and $\|\mathrm{Law}(y_l \mid C) - \nu_{x_l, p_l^C}\|_{TV} \le e_l + |\tilde{p}_l - \pi_l| + |\pi_l - p_l^C|$ (note $\|\nu_{x,p} - \nu_{x,p'}\|_{TV} = |p - p'|$). Inserting the bounds for o_l, $1/x_l$, $1/(1-x_l)$ and $\underline{N}_l \ge \min_C N_{k+1}$ gives the claim. ☐

Corollary 5.4 (Forgetting)**.** *Let C be a k-regular level-i cell and $C' \subset C$ a level-$(i+1)$ sub-cell, $i+2 \le k$. Using a common reference point $\alpha^* \in C'$,*

$$|p_k^C - p_k^{C'}| \le x_k \, 2^{-\lfloor (k-i-2)/2 \rfloor}.$$

Moreover $1 - p_l^C \le 1 - x_l$ for $i \le l \le k$, and $p_l^C \le 2x_l$ for $i+1 \le l \le k$; in particular ν_{x_l, p_l^C} has density at most 2 for $i+1 \le l \le k$.

Proof. p^C and $p^{C'}$ solve (5.2) with the same environment from level i on, with initial values in $[0,1]$ at level i, so the first claim is Lemma 5.2(iv). The others are Lemma 5.2(i)–(ii) and (5.3). ☐

Remark 5.5. Theorem 5.3 expresses the conditional law of y_k given a cell at level i through the *digits only*, up to the initial value p_{i-1}^C, whose influence decays like $2^{-(k-i)/2}$ *uniformly in the digits*. Section 4 and the present section are taken from [6], where the recursion is also checked numerically.

6. Facts on continued fractions

We use the following standard facts (see [11, 8]).

(CF1) Cylinders. For integers $b_1, \dots, b_m \ge 1$ with denominators q_t, $\lambda\{a_1 = b_1, \dots, a_m = b_m\} = \frac{1}{q_m(q_m + q_{m-1})} \le \prod_{t=1}^{m} b_t^{-2}$. The density h_j of $G_*^j \lambda$ satisfies $\frac{1}{2} \le h_j \le 2$: indeed $h_j(x) = \sum (q_j +$

$q_{j-1}x)^{-2}$, the sum running over cylinders of depth j, and $\frac{1}{2q_j(q_j+q_{j-1})} \le (q_j+q_{j-1}x)^{-2} \le \frac{2}{q_j(q_j+q_{j-1})}$ because $q_{j-1} \le q_j$.

(CF2) Mixing (Kuzmin–Lévy–Wirsing, ψ-mixing). This is the Gauss–Kuzmin–Wirsing theorem applied to the normalised restrictions of λ to cylinders, whose images under G^j have densities $\frac{1+u}{(1+ut)^2}$, $u = q_{j-1}/q_j \in [0,1]$, a family on which the convergence to the Gauss density is uniform and exponential [8, Ch. 2]. There are constants C_1 and $\rho \in (0,1)$ such that for all $j, n \ge 0$, all $A \in \sigma(a_1, \dots, a_j)$ and all Borel $E \subset (0,1)$,

$$\bigl|\lambda(A \cap G^{-(j+n)}E) - \lambda(A)\gamma(E)\bigr| \le C_1\rho^n\lambda(A)\gamma(E).$$

Since the densities h_j of $G_*^j\lambda$ lie in $[\frac{1}{2}, 2]$ (by the computation in (CF1)), it follows that there is a constant C_2 such that, if f is $\sigma(a_1, \dots, a_j)$-measurable and $g = \hat{g} \circ G^{j+n}$ for a Borel function $\hat{g}$, then

$$|\mathbb{E}[fg] - \mathbb{E}f\,\mathbb{E}g| \le C_2\rho^n\,\mathbb{E}|f|\,\mathbb{E}|g|, \qquad \Bigl|\mathbb{E}g - \int \hat{g}\,d\gamma\Bigr| \le C_1\rho^{j+n}\int |\hat{g}|\,d\gamma;$$

in particular $\mathbb{P}(a_{k+1} = b) = \frac{1+O(\rho^k)+O(1/b)}{b^2\log 2}$.

(CF3) Products of digits. For $0 < \delta \le \frac{1}{2}$, $j \ge 0$ and $m \ge 1$, $\mathbb{E}\prod_{t=j+1}^{j+m}(a_t+1)^{1-\delta} \le 2(4/\delta)^m$; hence $\mathbb{P}\bigl(\prod_{t=j+1}^{j+m}(a_t+1) > x\bigr) \le 2(4/\delta)^m x^{-1+\delta}$.

Proof. By (CF1) applied to $G_*^j\lambda$, the expectation is at most

$$2\Bigl(\sum_{b\ge1}(b+1)^{1-\delta}b^{-2}\Bigr)^m \le 2\Bigl(2\sum_{b} b^{-1-\delta}\Bigr)^m \le 2(4/\delta)^m.$$

□

(CF4) Growth of q_k. $\mathbb{E}\log q_k = \frac{\pi^2}{12\log 2}k + O(1)$ and $\operatorname{Var}(\log q_k) \le Ck$ (Lévy; for the variance bound write $\log q_k = \sum_{j<k}\log(1/\alpha_j) + O(1)$ by (2.1) and replace $\log(1/\alpha_j)$ by its conditional expectation given $a_{j+1}, \dots, a_{j+r}$, which changes it by $O(2^{-r})$; (CF2) then bounds the covariances. In fact $\log q_k$ satisfies a central limit theorem [8]). We write

$$c_0 = \frac{12\log 2}{\pi^2}.$$

(CF5) Large digits. $\mathbb{P}(a_{k+1} \ge b) \le 2/b$ for all $k \ge 0$, $b \ge 1$, since $h_k \le 2$.

(CF6) The next digit. For $k \ge 0$ and $b \ge 1$,

$$\mathbb{P}(a_{k+1} = b \mid \mathcal{F}_k) = p_{\omega_k}(b) := \frac{1+\omega_k}{(b+\omega_k)(b+1+\omega_k)}, \qquad \mathbb{P}(a_{k+1} > T \mid \mathcal{F}_k) = \frac{1+\omega_k}{\lfloor T\rfloor + 1 + \omega_k} \le \frac{2}{T},$$

where $\omega_k = q_{k-1}/q_k$, because conditionally on $\mathcal{F}_k$ the variable α_k has density $(1+\omega_k)(1+\omega_k t)^{-2}$ on $(0,1)$ (the case $\mathcal{I} = (0,1)$ of Lemma 3.3). In particular $|p_\omega(b) - (1+\omega)b^{-2}| \le 10b^{-3}$ and $|\partial_\omega p_\omega(b)| \le 3b^{-2}$.

Functions of the digits alone are handled by (CF2); functions which also involve the residue vectors w_k or the endpoint orbits are handled by the mixing of the extended chain, Lemmas 3.6 and 3.7.

7. Set-up and decomposition

7.1. **Parameters.** Let $N \to \infty$, $L = \log N$, and fix $\varepsilon \in (0, \frac{1}{2})$. Put

$$T = \varepsilon L, \quad U = L^{1.1}, \quad Z = L^{10}, \quad s = \lceil \log\log L \rceil, \quad M = \lceil 3L \rceil, \quad K_0 = \lfloor c_0(L - 10\log L) \rfloor.$$

For large N we have $J \le M$ (Lemma 2.2 and $q_j \ge 2^{(j-1)/2}$). We write $o_{\mathbb{P}}(1)$ for random variables tending to 0 in probability as $N \to \infty$ (ε fixed), and $\Pi_k = \prod_{u=k-s}^{k+2}(a_u + 1)$.

7.2. **Bulk indicators.** For $s < k \le M$ define the $\mathcal{H}_{k-s}$-measurable indicator

$$g_k = \mathbf{1}\{N_{k-s} \ge Z\} \cdot \mathbf{1}\{\mathcal{C}_{k-s}(\alpha) \text{ is } k\text{-regular}\} \cdot \mathbf{1}\{\Pi_k \le L^{1.2}\}$$

(on a k-regular level-$(k-s)$ cell the digits $a_{k-s}, \dots, a_{k+2}$ are constant, so g_k is a function of the cell), and split $g_k = \chi_k + \chi_k'$ with $\chi_k = g_k \mathbf{1}\{a_{k+1} \le T\}$, $\chi_k' = g_k \mathbf{1}\{a_{k+1} > T\}$; both are $\mathcal{H}_{k-s}$-measurable.

Lemma 7.1. *If $g_k = 1$ then $N_l \ge L^{8.7}$ for all $l \le k+1$, in particular $k+1 < J$, and every level-i cell $C \subset \mathcal{C}_{k-s}(\alpha)$ with $k - s \le i \le k$ is k-regular with $E_C \le 2L^{-2.7}$ (see (5.4)).*

Proof. $N_{l+1} \ge N_l\alpha_l - 1 \ge N_l/(a_{l+1}+1) - 1$, so $N_{k+1} \ge Z/\Pi_k - 2 \ge L^{8.7}$ and $N_l \ge N_{k+1}$ for $l \le k+1$. Sub-cells of a k-regular cell are k-regular (Section 4), and $E_C \le N_{k+1}^{-1} + \Pi_k^5 N_{i-1}^{-1} \le 2L^{6-8.7}$. □

Lemma 7.2 (Good event)**.** *Let G_N be the event that $a_t \le U$ for all $t \le M+3$, that $\Pi_k \le L^{1.2}$ for all $s < k \le M$, and that $\mathcal{C}_{k-s}(\alpha)$ is k-regular for every $s < k \le M$ with $N_{k-s} \ge Z$. Then $\mathbb{P}(G_N) \to 1$.*

Proof. By (CF5), $\mathbb{P}(\exists t \le M+3 : a_t > U) \le 2(M+3)/U \to 0$. By (CF3) with $\delta = \frac{1}{10}$ and $m = s+3$, $\mathbb{P}(\Pi_k > L^{1.2}) \le 2 \cdot 40^{s+3} L^{-1.08} = L^{-1.08+o(1)}$, and we sum over $k \le M$. For regularity, fix k with $N_{k-s} \ge Z$ and $\Pi_k \le L^{1.2}$, let $C = \mathcal{C}_{k-s}(\alpha)$ and let $D = D_{k+2}(\alpha)$ be the cylinder of depth $k+2$ containing α. *Case 1: $\overline{C} \subset \operatorname{int} D$.* Then $a_1, \dots, a_{k+2}$ are constant on a neighbourhood of $\overline{C}$; in particular Π_k is constant on C, so the bound $N_{l+1} \ge N_l/(a_{l+1}+1) - 1$ of Lemma 7.1 gives $N_l \ge L^{8.7} \ge 2$ on all of C for $l \le k+1$, and C is k-regular. *Case 2: $\overline{C} \not\subset \operatorname{int} D$.* Then $\operatorname{dist}(\alpha, \partial D) \le |C|$. By the proof of Lemma 4.6 and Lemma 2.2, $|C| \le (N_{k-s-1} q_{k-s-1}^2)^{-1} \le 4/(N q_{k-s-1})$, while $\lambda(D) \ge (2q_{k+2}^2)^{-1}$ and $q_{k+2} \le q_{k-s-1}\Pi_k$. Since $N_{k-s} \ge Z$ forces $q_{k-s-1} < q_{k-s} < N/Z$, we get $|C|/\lambda(D) \le 8\Pi_k^2/Z \le L^{-3}$. So Case 2 implies $\operatorname{dist}(\alpha, \partial D_{k+2}(\alpha)) \le L^{-3}\lambda(D_{k+2}(\alpha))$. Since the cylinders of depth $k+2$ partition $(0,1)$ up to a null set, $\lambda\{\alpha : \operatorname{dist}(\alpha, \partial D_{k+2}(\alpha)) \le \eta\lambda(D_{k+2}(\alpha))\} = 2\eta$ for $\eta \le \frac{1}{2}$; with $\eta = L^{-3}$ and a union bound over $k \le M$, the probability of non-regularity is $O(L^{-2})$. □

Lemma 7.3 (Bulk end)**.** *Let $K'' = \max\{l : N_l \ge Z\}$. Then $J - K'' \le 2\log_2(Z+4) + 4$, and for every $\eta > 0$ there is A such that $\mathbb{P}(|K'' - K_0| > A\sqrt{L}) \le \eta$ for all large N.*

Proof. By Lemma 2.2, $N|\theta_{K''}| < Z + 4$; for $K'' < k < J$ we need $N_k \ge 1$, i.e. $N|\theta_{k-1}| \ge 1$, and $|\theta_{k+1}| \le |\theta_{k-1}|/2$, which gives the first claim. For the second, $N_l \ge Z$ holds if $q_l \le N/(2Z+8)$ and fails if $q_l > N/Z$ (Lemma 2.2). Hence $m_1 \le K'' \le m_2$ with $m_1 = \max\{l : q_l \le N/(2Z+8)\}$ and $m_2 = \max\{l : q_l \le N/Z\}$. As q_l increases with l, $\{m_2 > K_0 + A\sqrt{L}\} \subset \{\log q_{l_+} \le L - 10\log L\}$ with $l_+ = K_0 + \lceil A\sqrt{L} \rceil$, and $\{m_1 < K_0 - A\sqrt{L}\} \subset \{\log q_{l_-} > L - 10\log L - \log(2 + 8/Z)\}$ with $l_- = K_0 - \lfloor A\sqrt{L} \rfloor$. By (CF4), $\mathbb{E}\log q_{l_\pm} = L - 10\log L \pm A\sqrt{L}/c_0 + O(1)$ and $\operatorname{Var}\log q_{l_\pm} = O(L)$, so Chebyshev's inequality bounds both probabilities by Cc_0^2/A^2, which is $\le \eta/2$ for A large. □

7.3. **The decomposition.** From now on $\beta, r \in \mathbb{Q}$ are fixed as in Theorem 1.1, with endpoints $z_1 = \{-\beta\}$, $z_2 = \{r - \beta\}$ and common denominator Q; constants may depend on Q (hence on β, r). If one of the endpoints is 0, all terms attached to it below are 0. For an endpoint $z_i \ne 0$

write $z_{i,k}$ for its orbit, $u_{i,k} = \{P_i(w_k)_2/Q\}$ with $P_i = Qz_i$, and $v_{i,k}$ as in Lemma 3.1; for $z_i = 0$ put $u_{i,k} = 0$. Writing $z' = \{-z/x\}$ and $\tilde\kappa = \kappa - z'$, i.e.

$$\tilde\kappa(x,y,z) = \begin{cases} -1 & y \ge z, \\ -(1-y') + \mathbf{1}\{y' < z'\} - z' & y < z, \end{cases}$$

we have $\ell(x,y,z) = \tau(y,z)/x + z' + \tilde\kappa(x,y,z)$. Since $y' - z' + \mathbf{1}\{y' < z'\} = \{y' - z'\}$ for $y', z' \in [0,1)$,

$$\tilde\kappa(x,y,z) = -1 + \mathbf{1}\{y < z\}\,\{y' - z'\} \in [-1, 0). \tag{7.1}$$

Hence $\tilde\kappa$ depends on (y', z') only through $y' - z'$ modulo 1; it is continuous in z' on the circle and in y at $y' = 0$, it is continuous at $y = z$ (as $y \uparrow z$, $y' \downarrow z'$ and $\{y' - z'\} \to 0$), and its only other discontinuities are jumps of size 1 where $y' = z'$ with $y < z$, or where the case $y < z$ changes because z passes through 0. Accordingly, by (2.4),

$$L_k = M_k + Z_k + K_k, \qquad M_k = \frac{\tau(y_k, z_{2,k}) - \tau(y_k, z_{1,k})}{\alpha_k}, \qquad Z_k = z_{2,k+1} - z_{1,k+1}, \tag{7.2}$$
$$K_k = \tilde\kappa(\alpha_k, y_k, z_{2,k}) - \tilde\kappa(\alpha_k, y_k, z_{1,k}),$$

with $|M_k| \le \frac{a_{k+1}+1}{4}$, $|Z_k| \le 1$, $|K_k| \le 1$. The variable Z_k depends on α only. We split

$$D_N(\alpha, \beta; r) = \Lambda + \Lambda_Z + H + R, \tag{7.3}$$

$$\Lambda = \sum_{k=s+1}^{M} (-1)^k \chi_k (M_k + K_k), \quad \Lambda_Z = \sum_{k=s+1}^{M} (-1)^k \chi_k Z_k, \quad H = \sum_{k=s+1}^{M} (-1)^k \chi'_k L_k,$$

and $R = \sum_{k<J} (-1)^k L_k (1 - g_k)$, with $g_k := 0$ for $k \le s$.

Lemma 7.4. *$R/L \to 0$ in probability.*

Proof. Recall $|L_k| \le 2(a_{k+1} + 1)$. The terms with $k \le s$ contribute at most $2\sum_{k \le s}(a_{k+1} + 1)$, which is $O_{\mathbb{P}}(s^2)$ by (CF5). On G_N, $g_k = \mathbf{1}\{N_{k-s} \ge Z\}$ for $s < k \le M$, so the remaining terms have $K'' + s < k < J$; by Lemma 7.3 there are at most $2\log_2(Z+4) + 4 = O(\log L)$ of them, and with probability $\ge 1 - \eta$ they lie in the deterministic window $W = [K_0 - A\sqrt{L},\ K_0 + A\sqrt{L} + C\log L]$. Hence, for $\delta > 0$,

$$\mathbb{P}(|R| > \delta L) \le \mathbb{P}(G_N^c) + \eta + \mathbb{P}\Big(\sum_{k \le s}(a_{k+1} + 1) > \tfrac{\delta L}{4}\Big) + \mathbb{P}\Big(\max_{k \in W} a_{k+1} > \tfrac{\delta L}{C' \log L}\Big),$$

and by (CF5) the last two terms are $O(s^2/(\delta L))$ and $O(\sqrt{L}\log L/(\delta L))$. □

Thus it remains to show that $(\Lambda + \Lambda_Z)/L$ is small in probability uniformly as $\varepsilon \to 0$ (Section 8) and that H/L converges to a compound Poisson law (Section 9).

8. The bounded part

Proposition 8.1. *There is a constant C_Q such that for every $\delta > 0$, $\limsup_{N\to\infty} \mathbb{P}(|\Lambda + \Lambda_Z| > \delta L) \le C_Q \varepsilon/\delta^2$.*

Throughout this section we work on $G_N \cap \{|K'' - K_0| \le A\sqrt{L}\}$, whose complement has probability $\le \eta + o(1)$ with $\eta > 0$ arbitrary (Lemmas 7.2 and 7.3); there $\chi_k = \mathbf{1}\{a_{k+1} \le T\}\mathbf{1}\{k \le K'' + s\}$ for $s < k \le M$. Let $W' = [K_0 - A\sqrt{L},\ K_0 + A\sqrt{L} + s]$, a deterministic window of the same kind as in Lemma 7.4.

8.1. **Approximation by finitely many pieces.** For $l \geq 0$ and $r \geq 1$ the pieces $\mathfrak{p}_l, \dots, \mathfrak{p}_{l+r}$ determine the digits $a_{l+1}, \dots, a_{l+r+1}$, so α_l is determined up to an interval of length at most F_{r+1}^{-2} (F_j the Fibonacci numbers). We write $\alpha_l^{[r]}$ for the midpoint of this interval; it is a function of $\mathfrak{p}_l, \dots, \mathfrak{p}_{l+r}$ and $|\alpha_l - \alpha_l^{[r]}| \leq F_{r+1}^{-2}$. Inside a fixed piece $\mathfrak{p}_l$ the integer parts $n_l^{(i)}$ are constant (Lemma 3.2(a)), so $z_{i,l} = u_{i,l} + v_{i,l}\alpha_l - n_l^{(i)}$ is an affine function of α_l with slope $|v_{i,l}| < 4$.

8.2. **The part depending on α only.** On the event above, $\Lambda_Z = \Sigma_Z + O(|W'|)$ with $\Sigma_Z = \sum_{s<k\leq K_0}(-1)^k \hat{z}_k$ and $\hat{z}_k = \mathbf{1}\{a_{k+1} \leq T\}Z_k$; note $|W'| = o(L)$.

Lemma 8.2. $\mathbb{E}[\Sigma_Z^2] \leq C_Q L$.

Proof. $\hat{z}_k = \mathbf{1}\{a(\mathfrak{p}_k) \leq T\}\big(z_{2,k+1} - z_{1,k+1}\big)$ is a functional of the chain from time $k+1$ together with $\mathfrak{p}_k$, i.e. of the chain from time k, the same functional for every k, bounded by 1 and independent of ω. By Lemma 3.6(b), $\mathbb{E}\hat{z}_k = \tilde{\mu} + O(\vartheta_Q'^k)$, so the alternating sum of the means is $O(1)$. Let $\hat{z}_k^{[r]}$ be obtained by replacing α_{k+1} by $\alpha_{k+1}^{[r]}$; then $|\hat{z}_k - \hat{z}_k^{[r]}| \leq 8F_{r+1}^{-2}$ and $\hat{z}_k^{[r]}$ is $\mathcal{F}'_{k+r+2}$-measurable. For $j < k$ and $r = \lfloor (k-j)/2 \rfloor$, Lemma 3.6(a) gives $|\mathrm{Cov}(\hat{z}_j, \hat{z}_k)| \leq |\mathrm{Cov}(\hat{z}_j^{[r]}, \hat{z}_k)| + 16F_{r+1}^{-2} \leq C_Q \vartheta_Q'^{k-j-r-2} + 16F_{r+1}^{-2}$, which is summable in $k-j$. Hence $\mathrm{Var}(\Sigma_Z) \leq C_Q L$. □

8.3. **Continuity properties.** Call an interval I of parameters *k-clean* if $a_1, \dots, a_{k+2}$ are constant on I and $\alpha_{k-1}(I)$, $\alpha_k(I)$ contain no point of E_0. We use that G maps $E_0 \setminus \{1/a : a \geq 1\}$ into E_0 (proof of Lemma 3.2(b)). Hence, on an interval on which $a_1, \dots, a_{k+2}$ are constant, a parameter at which α_j ($j \leq k$) passes through a point $e \in E_0$ is also a parameter at which $\alpha_{j+1}, \dots, \alpha_k$ pass through the points $G(e), \dots, G^{k-j}(e) \in E_0$: these are not branch points, since $\alpha_{j+1}, \dots, \alpha_k$ stay inside fixed open branches.

Lemma 8.3. *On a k-clean interval I the pieces $\mathfrak{p}_0, \dots, \mathfrak{p}_k$ are constant, hence so are ζ_k and ζ_{k+1}. Consequently $z_{i,k} = u_{i,k} + v_{i,k}\alpha_k - n_{i,k}$ is a continuous function on I with values in $(0,1)$, $z_{i,k+1} = \{u_{i,k+1} + v_{i,k+1}\alpha_{k+1}\}$ is continuous as a map to the circle $\mathbb{R}/\mathbb{Z}$, and $|dz_{i,l}/d\alpha| \leq 16q_l^2$ for $l = k, k+1$.*

Proof. By the remark above, a change of $\mathfrak{p}_j$ for some $j \leq k$ on I (which can only happen where α_j passes through a point of E_0, the digits being constant) would force α_k to pass through a point of E_0, which is excluded. So $\mathfrak{p}_0, \dots, \mathfrak{p}_k$ are constant and, by (3.2), so are ζ_k, ζ_{k+1}. The formulas follow from Lemma 3.1; $z_{i,k}$ does not vanish on I because $\alpha_k(I)$ contains no point of E_0. Finally $|dz_{i,l}/d\alpha| \leq |v_{i,l}|\,|d\alpha_l/d\alpha| \leq 4(q_l + q_{l-1}\alpha_l)^2 \leq 16q_l^2$. □

If $a_1, \dots, a_{k+2}$ are constant on an interval but it is not k-clean, consider a parameter at which some α_j passes through E_0. For each endpoint, at most one of the circle points $z_{i,l}$ jumps there: if $z_{i,j}$ passes through 0, then $z_{i,j+1}$ has the one-sided limits 1^- and $(1-\alpha_{j+1})^+$, both are mapped by $t \mapsto \{-t/\alpha_{j+1}\}$ to $1 - \alpha_{j+2}$, so $z_{i,j+2}$ is continuous with value $1 - \alpha_{j+2} \neq 0$, and inductively $z_{i,l} = 1 - \alpha_l \neq 0$ is continuous for $l \geq j+2$. In particular $z_{i,k}$ jumps by at most α_k and $z_{i,k+1}$ by at most α_{k+1} on the circle.

Let $G_k = M_k + K_k$. On a k-clean interval G_k is a continuous function of $(\alpha_k, y_k, y_{k+1}, z_{i,k}, z_{i,k+1})$, except at parameters where $y_k < z_{i,k}$ and $y_{k+1} = z_{i,k+1}$ for some i, where K_k jumps by 1: indeed τ is 1-Lipschitz on the circle in each variable, and $\tilde{\kappa}$ is continuous at $y = z$, at $y' = 0$ and in $z' \in \mathbb{R}/\mathbb{Z}$, and 1-Lipschitz in (y', z') elsewhere (7.1). Without the cleanness assumption, every discontinuity of G_k on an interval on which $a_1, \dots, a_{k+2}$ are constant is a jump of size at most 10: at a parameter where α_k passes through E_0, each $z_{i,k}$ moves by at most α_k on the circle, hence M_k by at most 2, and each $\tilde{\kappa} \in [-1, 0)$ changes by at most 1.

8.4. **Oscillation bounds.** For a level-i cell C let $\mathcal{B}_k(C)$ be the event that C is not k-clean or that $y_{k+1}-z_{i',k+1}$ vanishes (on the circle) at some point of C for some endpoint i'; it is a property of the cell.

Lemma 8.4. *Suppose $\chi_k=1$, $k+3\le i\le J$, and let $C=\mathcal{C}_i(\alpha)$. Then*

$$\operatorname{osc}_C G_k \le 6e^4\,2^{-\lfloor (i-k-2)/2\rfloor}+L^{-5}+90\,\mathbf{1}_{\mathcal{B}_k(C)}, \tag{8.1}$$

and

$$\mathbb{P}\big(\chi_k=1,\ i\le J,\ \mathcal{B}_k(\mathcal{C}_i(\alpha))\big)\le C\,2^{-\lfloor (i-k-2)/2\rfloor}+C_Q(s+2)L^{-2.7}. \tag{8.2}$$

The same bounds hold with $i=J$ and $2^{-\lfloor (i-k-2)/2\rfloor}$ replaced by L^{-8}.

Proof. On C, $a_1,\dots,a_{k+2}$ and $N_1,\dots,N_i$ are constant ($C\subset\mathcal{C}_{k+1}(\alpha)$, which is k-regular). By Lemma 4.7(a), $\operatorname{osc}_C y_k\le e^4a_{k+1}^{-1}2^{-\lfloor (i-k-2)/2\rfloor}$ and $\operatorname{osc}_C y_{k+1}\le e^4 2^{-\lfloor (i-k-2)/2\rfloor}$; by Lemma 4.7(b), $\operatorname{osc}_C\alpha_k\le 4/N_k$ and $\operatorname{osc}_C\alpha_{k-1}\le 4/N_{k-1}$. A level-$i$ cell has length $|C|\le (N_{i-1}q_{i-1}^2)^{-1}$ (proof of Lemma 4.6), and $N_{i-1}q_{i-1}\ge N/16$ by Lemma 2.2; hence on a k-clean cell, by Lemma 8.3, $\operatorname{osc}_C z_{i',l}\le 256q_{k+1}/N\le 256/N_{k+1}$ for $l\le k+1$. Since $|\partial_y\tau|,|\partial_z\tau|\le 1$, $\alpha_k^{-1}\le a_{k+1}+1\le T+1$, $\tau\le\frac14$, and $N_l\ge L^{8.7}$ for $l\le k+1$ (Lemma 7.1),

$$\operatorname{osc}_C M_k\le 2(a_{k+1}+1)\operatorname{osc}_C y_k+(T+1)\sum_{i'}\operatorname{osc}_C z_{i',k}+(T+1)^2\operatorname{osc}_C\alpha_k\le 4e^4\,2^{-\lfloor (i-k-2)/2\rfloor}+\tfrac12 L^{-5},$$

and, if moreover $y_{k+1}-z_{i',k+1}$ has no zero on C, $\operatorname{osc}_C K_k\le 2\operatorname{osc}_C y_{k+1}+2\sum_{i'}\operatorname{osc}_C z_{i',k+1}\le 2e^4\,2^{-\lfloor (i-k-2)/2\rfloor}+\frac12L^{-5}$. Otherwise G_k has at most 9 discontinuities on C. Indeed, $\alpha_k(C)$ has length $<D_0^{-2}$, so it contains at most one point of E_0 (distinct points of E_0 are at distance $\ge D_0^{-2}$), and by the remark before Lemma 8.3 every parameter at which some α_j, $j\le k$, passes through E_0 is one at which α_k does; so there is at most one such parameter. On each of the (at most two) remaining subintervals ζ_{k+1} is constant, and a lift of $y_{k+1}-z_{i',k+1}$ is strictly increasing as a function of α_{k+1} ($y_{k+1}=N_{k+1}\alpha_{k+1}-N_{k+2}$ increases, and $z_{i',k+1}$ is affine in α_{k+1} with slope $v_{i',k+1}\le 0$), with total increase $<1+256/N_{k+1}<2$; so it passes through an integer at most twice. This gives at most $1+2\cdot2\cdot2$ discontinuities, each a jump of size at most 10, which gives (8.1).

For (8.2): if C is not k-clean, then α_{k-1} or α_k is within $4L^{-8.7}$ of E_0 at the point α; since the density of α_l is at most 2 (CF1), this has probability $\le C_QL^{-8.7}$. Otherwise a zero of $y_{k+1}-z_{i',k+1}$ on C forces $\|y_{k+1}-z_{i',k+1}\|\le\delta:=e^4 2^{-\lfloor (i-k-3)/2\rfloor}+256L^{-8.7}$ at α. Condition on $C'=\mathcal{C}_{k-s}(\alpha)$, which is k-regular with $E_{C'}\le 2L^{-2.7}$ (Lemma 7.1), and fix a reference point $\alpha^*\in C'$. By Theorem 5.3, $\operatorname{Law}(y_k\mid C')$ is within total variation $C_0(s+2)E_{C'}$ of $\nu_{x_k,p}$, $p=p_k^{C'}$. On C', $\operatorname{osc}\alpha_k\le 4\Pi_k^2/N_{k-s-1}\le 4L^{-7.6}$ (Lemma 4.7(b)), so $\|y_{k+1}-T_{x_k}(y_k)\|\le 2\cdot4L^{-7.6}(T+1)^2\le L^{-5}$; by Lemma 5.1, T_{x_k} maps $\nu_{x_k,p}$ to $\nu_{x_{k+1},p'}$ with $p'=\lambda(x_k)p+\mu(x_k)$, whose density is at most 2 (Corollary 5.4, the recursion being extended to level $k+1$ with the digit a_{k+2}, constant on C'). Finally, off an event of probability $\le C_QL^{-7.6}$ (on which C' is not k-clean), $z_{i',k+1}$ varies on C' by at most $16q_{k+1}^2|C'|\le 256\Pi_k/N_{k+1}\le L^{-7}$. Hence $\mathbb{P}(\|y_{k+1}-z_{i',k+1}\|\le\delta\mid C')\le 4(\delta+2L^{-5})+C_0(s+2)E_{C'}$, and (8.2) follows. For $i=J$ use that $\chi_k=1$ forces $N_{k+1}\ge L^{8.7}$, hence $J-k\ge 17\log_2 L-2$ (as $N_{l+2}<N_l/2$), so all the factors $2^{-\lfloor (i-k-2)/2\rfloor}$ above are at most L^{-8}. □

8.5. **The martingale decomposition.** Put $X_k=(-1)^k\chi_kG_k$ for $s<k\le M$, so $\Lambda=\sum_k X_k$, and $\Delta_{k,i}=\mathbb{E}[X_k\mid\mathcal{H}_{i+1}]-\mathbb{E}[X_k\mid\mathcal{H}_i]$. For the $(\mathcal{H}_i)$-stopping time $J\le M$ put $\mathbb{E}[X_k\mid\mathcal{H}_J]=\sum_j\mathbf{1}\{J=j\}\mathbb{E}[X_k\mid\mathcal{H}_j]$. Since χ_k is $\mathcal{H}_{k-s}$-measurable and $\chi_k=1$ forces $J>k+1$, on $\{J\le$

$k-s\} \in \mathcal{H}_{k-s}$ all the conditional expectations below vanish, while on $\{J > k-s\}$ the telescoping of the martingale $i \mapsto \mathbb{E}[X_k \mid \mathcal{H}_i]$ between $k-s$ and J gives, pointwise,

$$X_k = \mathbb{E}[X_k \mid \mathcal{H}_{k-s}] + \sum_{i=k-s}^{M-1} \mathbf{1}\{i < J\}\Delta_{k,i} + \big(X_k - \mathbb{E}[X_k \mid \mathcal{H}_J]\big).$$

Summing over k we obtain $\Lambda = \Lambda_{\mathrm{res}} + \Lambda_{\mathrm{mart}} + \Lambda_{\mathrm{tail}}$ with

$$\Lambda_{\mathrm{res}} = \sum_k \mathbb{E}[X_k \mid \mathcal{H}_{k-s}], \qquad \Lambda_{\mathrm{mart}} = \sum_{i<M} \mathbf{1}\{i < J\}D_i, \quad D_i = \sum_{s<k\le \min(M,i+s)} \Delta_{k,i}.$$

For a k-regular level-i cell C ($k-s \le i \le k$) with reference point $\alpha^* \in C$, let

$$\bar G(x,p,z_1,z_2) = \int_0^1 G(x,u,z_1,z_2)\,\nu_{x,p}(du),$$

$$G(x,u,z_1,z_2) = \frac{\tau(u,z_2)-\tau(u,z_1)}{x} + \tilde\kappa(x,u,z_2) - \tilde\kappa(x,u,z_1),$$

where inside $\tilde\kappa$ we use $u' = \{-u/x\}$ and $z_i' = \{-z_i/x\}$. Then $G_k = G(\alpha_k, y_k, z_{1,k}, z_{2,k})$ and $\sup_u |G(x,u,\cdot)| \le \frac{a+1}{4} + 4$ with $a = \lfloor 1/x \rfloor$; by the remark after (5.3),

$$\Big|\frac{\partial \bar G}{\partial p}\Big| \le 2(a+5). \tag{8.3}$$

Lemma 8.5. *If $\chi_k = 1$ and $C = \mathcal{C}_i(\alpha)$ with $k-s \le i \le k$, then, with the reference point $\alpha^* = \alpha$,*

$$\big|\mathbb{E}[G_k \mid C] - \bar G\big(\alpha_k, p_k^C, z_{1,k}, z_{2,k}\big)\big| \le C(T+5)(s+2)L^{-2.7} + 20\cdot\mathbf{1}\{C \text{ is not } k\text{-clean}\},$$

and $\mathbb{P}(\chi_k = 1,\ \mathcal{C}_{k-s}(\alpha)$ is not k-clean$) \le C_Q L^{-7.6}$.

Proof. By Theorem 5.3, replacing $\mathrm{Law}(y_k \mid C)$ by ν_{x_k,p_k^C} costs at most $2(\frac{T+1}{4}+4)C_0(s+2)E_C$. On C the parameters α_k and (if C is k-clean) $z_{i',k}$, $z_{i',k+1}$ vary by at most $4L^{-7.6}$ and L^{-7} (proof of Lemma 8.4). For fixed u, the map $(x,z_1,z_2) \mapsto G(x,u,z_1,z_2)$ is Lipschitz with constant $\le 4(T+1)^2$ except where $u' = z_i'$ (with $u < z_i$) or where the case $u \ge z_i$ changes at $z_i = 0$; the latter does not happen on a clean cell, and the set of u for which $u' - z_i'$ changes sign when the parameters move by η has Lebesgue measure $\le 4\eta(T+1)^2$, while the density of ν_{x_k,p_k^C} is at most 2 (Corollary 5.4). This gives the bound on clean cells; on other cells the same argument applies on each side of the (at most two) crossings of E_0, and the values on the two sides differ by at most 10. The probability bound is the first part of the proof of (8.2). □

Lemma 8.6. $\mathbb{E}|\Lambda_{\mathrm{tail}}| = o(L)$.

Proof. $|X_k - \mathbb{E}[X_k \mid \mathcal{H}_J]| \le \mathrm{osc}_{\mathcal{C}_J(\alpha)} X_k$; apply Lemma 8.4 with $i = J$ and sum over $k \le M$: $\mathbb{E}|\Lambda_{\mathrm{tail}}| \le M(6e^4L^{-8} + L^{-5}) + 90\sum_k \mathbb{P}(\chi_k = 1, \mathcal{B}_k(\mathcal{C}_J(\alpha))) = O(L\cdot sL^{-2.7})$. □

Lemma 8.7. $\mathbb{E}[\Lambda_{\mathrm{mart}}^2] \le C\varepsilon L^2 + o(L^2)$.

Proof. The summands $\mathbf{1}\{i < J\}D_i$ are $\mathcal{H}_{i+1}$-measurable with $\mathbb{E}[\mathbf{1}\{i < J\}D_i \mid \mathcal{H}_i] = 0$, hence orthogonal, and $\mathbb{E}[\Lambda_{\mathrm{mart}}^2] = \sum_i \mathbb{E}[\mathbf{1}\{i < J\}D_i^2]$. We bound $\|\mathbf{1}\{i < J\}\Delta_{k,i}\|_2$, recalling that $\Delta_{k,i} = 0$ where $\chi_k = 0$. (a) If $i \ge k+3$, $|\Delta_{k,i}| \le \mathrm{osc}_{\mathcal{C}_i(\alpha)} X_k$, and by Lemma 8.4, $\|\mathbf{1}\{i < J\}\Delta_{k,i}\|_2 \le 6e^4 2^{-\lfloor (i-k-2)/2\rfloor} + L^{-5} + 90\big(C2^{-\lfloor (i-k-2)/2\rfloor} + C_Q(s+2)L^{-2.7}\big)^{1/2}$. (b) If $k-1 \le i \le k+2$, $|\Delta_{k,i}| \le 2(a_{k+1}+5)\mathbf{1}\{a_{k+1} \le T\}$, whose L^2 norm is at most $C\sqrt{T}$ because $\mathbb{P}(a_{k+1} = b) \le 2/b^2$. (c) If $k-s \le i \le k-2$, the cells $C = \mathcal{C}_i(\alpha)$ and $C' = \mathcal{C}_{i+1}(\alpha)$ are k-regular with $E_C, E_{C'} \le 2L^{-2.7}$. Lemma 8.5 applied to both (common reference point α), (8.3), Corollary 5.4 and $(a_{k+1}+5)\alpha_k \le 6$ give

$$|\Delta_{k,i}| \le 12\cdot 2^{-\lfloor (k-i-2)/2\rfloor} + C(T+5)(s+2)L^{-2.7} + 40\cdot\mathbf{1}\{\mathcal{C}_{k-s}(\alpha) \text{ is not } k\text{-clean}\}.$$

Summing, $\|\mathbf{1}\{i<J\}D_i\|_2 \le C\sqrt{T}+C+o(1)$, and there are at most $M\le 4L$ indices i. □

8.6. **The residual.** For $\chi_k=1$, Lemma 8.5 with $C=\mathcal{C}_{k-s}(\alpha)$ gives

$$\mathbb{E}[X_k \mid \mathcal{H}_{k-s}] = (-1)^k\chi_k\bar{G}(\alpha_k,p_k^C,z_{1,k},z_{2,k})+e_k, \qquad \textstyle\sum_k \mathbb{E}|e_k| = o(L).$$

Let $p_k^\natural$ be the solution of (5.2) along the orbit of α ($x_l=\alpha_l$) started at level $k-s-1$ from $p_{k-s-1}^\natural=\alpha_{k-s-1}$. By Lemma 5.2(iv), $|p_k^C-p_k^\natural|\le \alpha_k 2^{-\lfloor (s-1)/2\rfloor}$, so by (8.3) replacing p_k^C by $p_k^\natural$ changes each term by at most $12\cdot 2^{-\lfloor(s-1)/2\rfloor}$, in total $O(L2^{-s/2})=o(L)$.

Next, if $p\le 2x$ then $\|\nu_{x,p}-\nu_{x,x}\|_{TV}\le x$, so with $\bar\tau(z)=\int_0^1\tau(u,z)\,du=\frac{z(1-z)}{2}$ and $a=\lfloor 1/x\rfloor$,

$$\big|\bar{G}(x,p,z_1,z_2)-a\big(\bar\tau(z_2)-\bar\tau(z_1)\big)\big| \le \tfrac12+\tfrac14+4\le 5. \tag{8.4}$$

Moreover $\bar\tau$ is $\frac12$-Lipschitz on the circle and $\|z_{i,k}-u_{i,k}\|\le |v_{i,k}|\alpha_k<4\alpha_k$ (Lemma 3.1), so $a_{k+1}|\bar\tau(z_{i,k})-\bar\tau(u_{i,k})|\le 2$. Since $p_k^\natural\le 2\alpha_k$ (Lemma 5.2(ii)), we may write

$$\mathbf{1}\{a_{k+1}\le T\}\bar{G}(\alpha_k,p_k^\natural,z_{1,k},z_{2,k}) = f_k+h_k, \qquad f_k=\mathbf{1}\{a_{k+1}\le T\}\,a_{k+1}\,g(w_k), \quad |h_k|\le 9,$$

with $g(w)=\bar\tau(\{P_2w_2/Q\})-\bar\tau(\{P_1w_2/Q\})$. As in Lemma 7.4, on our event

$$\Big|\Lambda_{\mathrm{res}}-\Sigma_1-\Sigma_2\Big| \le \sum_{k\in W'}\mathbf{1}\{a_{k+1}\le T\}(a_{k+1}+9)+\sum_k|e_k|+o(L),$$

$$\Sigma_1=\sum_{s<k\le K_0}(-1)^k f_k, \qquad \Sigma_2=\sum_{s<k\le K_0}(-1)^k h_k,$$

and the window term has expectation $O((A\sqrt{L}+s)\log L)=o(L)$.

Lemma 8.8. *$\mathbb{E}[\Sigma_1^2]\le C_Q(\varepsilon L^2+L\log^2 L)$, and $\Sigma_2=o_{\mathbb{P}}(L)$; more precisely $\Sigma_2=\Sigma_2'+\Sigma_2''$ with $\mathbb{E}[\Sigma_2'^2]\le C_QL\log L$ and $|\Sigma_2''|\le L^{-1}$ on G_N.*

Proof. Σ_1. By (CF6), $\mathbb{E}[f_k\mid\mathcal{F}_k]=m_T(\omega_k)g(w_k)$ with $m_T(\omega)=\sum_{b\le T}b\,p_\omega(b)\le 2(1+\log T)$ and $|m_T'(\omega)|\le 3(1+\log T)$. The martingale differences $f_k-\mathbb{E}[f_k\mid\mathcal{F}_k]$ are orthogonal, and $\sum_k\mathbb{E}f_k^2\le K_0\cdot 2T\le C\varepsilon L^2$. The function $(\omega,w)\mapsto m_T(\omega)g(w)/(2+2\log T)$ is a 2-regular functional of the chain at time k (w_k is a function of ζ_k). By Lemma 3.6(b) its means converge exponentially fast, so $|\sum_k(-1)^k\mathbb{E}[m_T(\omega_k)g(w_k)]|\le C_Q\log T$, and by Lemma 3.6(a) its covariances at lag n are $O_Q(\vartheta_Q'^n\log^2 T)$. Hence $\mathbb{E}[(\sum_k(-1)^k m_T(\omega_k)g(w_k))^2]\le C_QL\log^2 L$.

Σ_2. The variable h_k is determined by ζ_{k-s-1} and the pieces $\mathfrak{p}_{k-s-1},\mathfrak{p}_{k-s},\dots$: the pieces determine the digits and the α_l, $p_k^\natural$ is computed from $\alpha_{k-s-1},\dots,\alpha_k$ and the digits, and $z_{i,k}=\{u_{i,k}+v_{i,k}\alpha_k\}$ with ζ_k obtained from ζ_{k-s-1} by (3.2). So $h_k=\phi(\zeta_{k-s-1},\mathfrak{p}_{k-s-1},\mathfrak{p}_{k-s},\dots)$ for a Borel function ϕ (depending on N only through s,T) with $|\phi|\le 9$, independent of ω. By Lemma 3.6(b) the alternating sum of the means $\mathbb{E}h_k$ is $O_Q(1)$. Let $r=\lceil 20\log L\rceil$ and let $h_k^{[r]}$ be obtained by replacing every α_l ($k-s-1\le l\le k+1$) by $\alpha_l^{[r+k+1-l]}$ in the formula for h_k and then clipping the result to $[-9,9]$ (the substituted values need not form a Gauss orbit, so the bound $|h_k|\le 9$ is not automatic); this is a function of ζ_{k-s-1} and $\mathfrak{p}_{k-s-1},\dots,\mathfrak{p}_{k+r+1}$, bounded by 9. Inside fixed pieces, ϕ is Lipschitz in these α_l with constant at most L^{10} on G_N: the maps $\lambda(\cdot),\mu(\cdot)$ of (5.1) have derivative $O(x^{-2})=O(U^2)$, $|\partial_p\bar G|\le 2(T+5)$, the density of $\nu_{x,p}$ depends on x with derivative $O(TU)$, the integrals of the $\tilde\kappa$-terms over u are Lipschitz in (x,z_i) with constant $O((T+1)^2)$ (the jumps of $\tilde\kappa$ in u only move), and $z_{i,l}$ is affine in α_l inside a piece. Since clipping is 1-Lipschitz and $|h_k|\le 9$, $|h_k-h_k^{[r]}|\le L^{10}F_{r+1}^{-2}\le L^{-5}$ on G_N. Put $\Sigma_2'=\sum_k(-1)^k h_k^{[r]}$ and $\Sigma_2''=\Sigma_2-\Sigma_2'$. The alternating sum of the means of $h_k^{[r]}$ is $O_Q(1)$ (Lemma 3.6(b)), and for $k-j>s+r+3$, Lemma 3.6(a) gives $|\mathrm{Cov}(h_j^{[r]},h_k^{[r]})|\le C_Q\vartheta_Q'^{k-j-s-r-3}$. Hence $\mathbb{E}[\Sigma_2'^2]\le C_QL(s+r)$. □

Proof of Proposition 8.1. Combine $\Lambda = \Lambda_{\mathrm{res}} + \Lambda_{\mathrm{mart}} + \Lambda_{\mathrm{tail}}$ with Lemmas 7.2, 7.3, 8.2, 8.6, 8.7, 8.8 and Chebyshev's inequality; η is arbitrary. □

9. The large partial quotients

For $c \in \mathbb{Z}/Q\mathbb{Z}$ let $m(t,c) = \tau(t,\{P_2 c/Q\}) - \tau(t,\{P_1 c/Q\})$ (with the convention $\tau(t,\{P_i c/Q\}) = 0$ for a zero endpoint), so that $|m| \le \frac14$, and let

$$\varphi_\varepsilon(\theta) = \exp\Big(\frac{c_0}{\log 2}\,\mathbb{E}\int_\varepsilon^\infty\!\!\int_0^1 \big(\cos(\theta x\, m(t,c_Q)) - 1\big)\,dt\,\frac{dx}{x^2}\Big),$$

the characteristic function of a symmetric compound Poisson law Π_ε.

Proposition 9.1. *H/L converges in distribution to Π_ε as $N \to \infty$.*

Reductions. Put $m_k(t) = m(t,(w_k)_2)$; thus $m_k(t) = \tau(t,u_{2,k}) - \tau(t,u_{1,k})$, and m_k is $\mathcal{F}_k$-measurable. By (7.2) and $\|z_{i,k} - u_{i,k}\| < 4\alpha_k$,

$$|L_k - a_{k+1} m_k(y_k)| \le \alpha_{k+1}\cdot\tfrac14 + a_{k+1}\sum_i \|z_{i,k} - u_{i,k}\| + |Z_k| + |K_k| \le 14.$$

Since $\mathbb{E}\#\{k \le M : a_{k+1} > T\} \le 2M/T \le 7/\varepsilon$, we have $|H - \sum_k (-1)^k \chi'_k a_{k+1} m_k(y_k)| \le 14\#\{k \le M : a_{k+1} > T\}$. Fix $\eta > 0$, $A = A(\eta)$ as in Lemma 7.3, put $K_* = K_0 - \lceil A\sqrt{L}\rceil$ and

$$\xi_k = \mathbf{1}\{T < a_{k+1} \le U\}\,\mathbf{1}\{s < k \le K_*\}, \qquad H_2 = \sum_k (-1)^k \xi_k a_{k+1} m_k(y_k).$$

On $G_N \cap \{|K'' - K_0| \le A\sqrt{L}\}$ we have $\chi'_k = \mathbf{1}\{a_{k+1} > T\}\mathbf{1}\{s < k \le K'' + s\}$, so $\chi'_k \ne \xi_k$ forces a partial quotient $a_{k+1} > T$ with $|k - K_0| \le A\sqrt{L} + s + 1$, an event of probability $O(\sqrt{L}/T)$. Hence, off an event of probability $\eta + o(1)$, $|H - H_2| \le 14\#\{k \le M : a_{k+1} > T\}$, whose expectation is $O(1/\varepsilon)$. Consequently

$$\limsup_{N\to\infty} \big|\mathbb{E}e^{i\theta H/L} - \varphi_\varepsilon(\theta)\big| \le 2\eta + \limsup_{N\to\infty} \big|\mathbb{E}e^{i\theta H_2/L} - \varphi_\varepsilon(\theta)\big|, \tag{9.1}$$

and it suffices to show that the last lim sup is at most $C(\theta)\sqrt{\eta}/\varepsilon$; since the left side of (9.1) does not depend on η, letting $\eta \to 0$ then proves Proposition 9.1. Put

$$\Theta_k = e^{i\theta(-1)^k a_{k+1} m_k(y_k)/L} - 1, \qquad \hat\Theta_k = \int_0^1 \big(e^{i\theta(-1)^k a_{k+1} m_k(t)/L} - 1\big)\,dt,$$

$A_k = 1 + \xi_k\Theta_k$ and $B_k = 1 + \xi_k\hat\Theta_k$; note $|A_k|, |B_k| \le 1$, $e^{i\theta H_2/L} = \prod_k A_k$, and B_k is $\mathcal{F}_{k+1}$-measurable.

Lemma 9.2 (Poisson limit for the digits and residues)**.** *$\mathbb{E}\prod_k B_k \to \varphi_\varepsilon(\theta)$.*

Proof. Let $\Gamma_k = \mathbb{E}[B_k \mid \mathcal{F}_k]$. By (CF6), $|\Gamma_k - 1| \le 2\mathbb{P}(a_{k+1} > T \mid \mathcal{F}_k) \le 4/T$, so $|\Gamma_k| \ge \frac12$ and $\prod_k |\Gamma_k|^{-1} \le e^{8M/T} \le e^{24/\varepsilon}$. Since $\mathbb{E}[B_k/\Gamma_k \mid \mathcal{F}_k] = 1$, the product $\prod_{k\le n} B_k/\Gamma_k$ has expectation 1 for every n, so

$$\Big|\mathbb{E}\prod_k B_k - \varphi_\varepsilon(\theta)\Big| = \Big|\mathbb{E}\Big[\prod_k \frac{B_k}{\Gamma_k}\Big(\prod_k \Gamma_k - \varphi_\varepsilon(\theta)\Big)\Big]\Big| \le e^{24/\varepsilon}\,\mathbb{E}\Big|\prod_k \Gamma_k - \varphi_\varepsilon(\theta)\Big|,$$

and it suffices to show $\prod_k \Gamma_k \to \varphi_\varepsilon(\theta)$ in probability (all quantities being bounded). As $\sum_k |\Gamma_k - 1|^2 \le 16M/T^2 \to 0$, it suffices to show $\sum_k(\Gamma_k - 1) \to \log\varphi_\varepsilon(\theta)$ in probability. By (CF6),

$$\Gamma_k - 1 = \xi_k^{(0)} \sum_{T < b \le U} p_{\omega_k}(b)\int_0^1 \big(e^{i\theta(-1)^k b m_k(t)/L} - 1\big)dt, \qquad \xi_k^{(0)} = \mathbf{1}\{s < k \le K_*\},$$

and replacing $p_{\omega_k}(b)$ by $(1+\omega_k)b^{-2}$ costs $\sum_{b>T} 20b^{-3} \le 20T^{-2}$ per k. A Riemann sum with $x = b/L$ (the integrand has derivative $O((|\theta|+\varepsilon^{-1})x^{-2})$ on $[\varepsilon,\infty)$) and the bound $\int_{U/L}^{\infty} 2x^{-2}dx = 2L/U$ give

$$\sum_k (\Gamma_k - 1) = \frac{1}{L}\sum_{s<k\le K_*} (1+\omega_k)\Big(R(c_k) + i(-1)^k I(c_k)\Big) + o(1),$$

where $c_k = (w_k)_2$ and, for $c \in \mathbb{Z}/Q\mathbb{Z}$,

$$R(c) = \int_\varepsilon^\infty \int_0^1 \big(\cos(\theta x m(t,c)) - 1\big)dt\,\frac{dx}{x^2}, \qquad I(c) = \int_\varepsilon^\infty \int_0^1 \sin(\theta x m(t,c))\,dt\,\frac{dx}{x^2};$$

note $|R|, |I| \le 2/\varepsilon$. The functionals $(1+\omega_k)R(c_k)\varepsilon/4$ and $(1+\omega_k)I(c_k)\varepsilon/4$ of the chain at time k are 1-regular. By Lemma 3.7, $\mathbb{E}[(1+\omega_k)R(c_k)] = \frac{1}{\log 2}\mathbb{E}R(c_Q) + O(\vartheta_Q'^k/\varepsilon)$, because $\int (1+t)\gamma(dt) = 1/\log 2$; and by Lemma 3.6(a) the covariances of these functionals at lag n are $O(\vartheta_Q'^n/\varepsilon^2)$. Hence, in L^2,

$$\frac{1}{L}\sum_{s<k\le K_*} (1+\omega_k)R(c_k) \to \frac{c_0}{\log 2}\mathbb{E}R(c_Q), \qquad \frac{1}{L}\sum_{s<k\le K_*} (-1)^k(1+\omega_k)I(c_k) \to 0,$$

using $(K_* - s)/L \to c_0$. Since $\log \varphi_\varepsilon(\theta) = \frac{c_0}{\log 2}\mathbb{E}R(c_Q)$, the lemma follows. □

It remains to show $\mathbb{E}\prod_k A_k - \mathbb{E}\prod_k B_k \to 0$. By telescoping,

$$\Big|\mathbb{E}\prod_k A_k - \mathbb{E}\prod_k B_k\Big| \le \sum_{k'} \Big|\mathbb{E}\Big[\prod_{k<k'} A_k \cdot \xi_{k'}(\Theta_{k'} - \hat\Theta_{k'}) \cdot \prod_{k>k'} B_k\Big]\Big|. \tag{9.2}$$

The difficulty is that $y_{k'}$ is *not* independent of the later digits that enter $\prod_{k>k'} B_k$: the cells reveal digits far ahead. We separate the later indices into those which are already determined by the cell $\mathcal{C}_{k'-s}(\alpha)$ and those which are still "mixing" given a cylinder on which $y_{k'}$ is essentially constant.

Near, far and transition indices. Fix k' with $\xi_{k'} = 1$ and put $X = Nq_{k'}$. Let

$$t_n = \max\{t \ge k' :\ q_{t+2}^2 \le L^{-6}X\}, \qquad t_f = \min\{t :\ q_t^2 \ge L^4 X\},$$
$$r_h = \lceil \log^2 L\rceil, \qquad w' = \lceil 10\log_2 L\rceil + 6 + r_h,$$

and call $k > k'$ *near* if $k \le t_n + 1$, *transition* if $t_n + 1 < k < t_n + w'$, and *far* if $k \ge t_n + w'$. Here $\{t_n = t\} \in \sigma(a_1,\dots,a_{t+3})$ and t_f is a stopping time of the digit filtration. Since $q_{t+2} \ge 2q_t$, the definitions give

$$t_n + 3 \le t_f \le t_n + w' - r_h. \tag{9.3}$$

Write $\prod_{k>k'} B_k = F_{\rm near}F_{\rm tr}F_{\rm far}$ accordingly, and $\prod_{k<k'} A_k = P'P_{\rm np}$ where $P_{\rm np}$ collects the indices $k' - 2s - 2 \le k < k'$.

Lemma 9.3 (Exceptional events)**.** *Let $\mathcal{E}_{k'}$ be the union of the events: (i) G_N^c, or $|K'' - K_0| > A\sqrt{L}$, or $g_{k'} = 0$; (ii) $\xi_k = 1$ for some $k \ne k'$ with $|k - k'| \le 2s + 2$; (iii) the set defining t_n is empty, or $\xi_k = 1$ for some transition index k; (iv) $\mathcal{C}_{k'-s}(\alpha) \not\subset D_{t_n+3}(\alpha)$, the cylinder of depth $t_n + 3$ containing α; (v) $D_{t_f}(\alpha) \not\subset \mathcal{C}_{k'+1}(\alpha)$. Then $\sum_{k'} \mathbb{E}[\xi_{k'}\mathbf{1}_{\mathcal{E}_{k'}}] \le C\sqrt{\eta}/\varepsilon + o(1)$.*

Proof. This involves the digits and the cells only, and the proof of [6, Lemma 8.3] applies verbatim; we recall it. (i) On $G_N \cap \{|K'' - K_0| \le A\sqrt{L}\}$ every $k' \le K_*$ has $g_{k'} = 1$; and $\mathbb{E}[\#\{k : \xi_k = 1\}^2] \le C/\varepsilon^2$ by (CF2), so the contribution is at most $(C/\varepsilon)(\eta + o(1))^{1/2}$. (ii) $\sum_{k'} \mathbb{P}(\xi_{k'} = \xi_k = 1,\ 0 < |k' - k| \le 2s + 2) \le CMs/T^2 \to 0$ by (CF2). (iii) Assume (i) fails. Then $k' \le K_* \le K''$, so $N_{k'} \ge Z$ and $q_{k'} < N/N_{k'} \le N/Z$ (Lemma 2.2). As $q_{k'+2} \le (a_{k'+1}+1)(a_{k'+2}+1)q_{k'}$ and $a_{k'+1} \le U$, emptiness forces $a_{k'+2} > L^{0.9}/2$, which has probability $O(L^{-0.9})$ given $\xi_{k'} = 1$. Transition indices $k \ge t_n + 3$ have a_{k+1} beyond the digits determining $\{t_n = t\} \cap \{\xi_{k'} = 1\}$, so by (CF2) each is heavy with conditional probability $\le C/T$, and there are at most w' of

them. For $k = t_n + 2$, the event $\{t_n = t,\ \xi_{t+2} = 1\}$ (which entails $T < a_{t+3} \le U$) forces $q_{t+2}^2 \in (L^{-6}X(U+1)^{-2}, L^{-6}X]$, which happens for at most $4\log_2(U+1)+2$ values of t, each time with conditional probability $\le 2/T$ given $\sigma(a_1,\dots,a_{t+2})$. Altogether this gives $C(w'+\log L)/T$ times $\mathbb{P}(\xi_{k'}=1) \le 2/T$, and the sum over $k' \le M$ is $O(L\log^2 L/T^2) \to 0$. (iv) Assume (i)–(iii) fail. Since $q_{t_n+2}^2 \le L^{-6}Nq_{k'} < L^{-16}N^2$, we have $t_n + 2 < m \le M$ (with m as in Lemma 2.2), so $a_{t_n+3} \le U$ on G_N. Further $|\mathcal{C}_{k'-s}| \le 4/(Nq_{k'-s-1})$ (proof of Lemma 7.2), $\lambda(D_{t_n+3}) \ge (2(U+1)^2 q_{t_n+2}^2)^{-1} \ge L^6/(2(U+1)^2X)$ and $q_{k'} \le q_{k'-s-1}\Pi_{k'}$, so $|\mathcal{C}_{k'-s}| \le L^{-2.5}\lambda(D_{t_n+3})$. Hence (iv) implies $\operatorname{dist}(\alpha, \partial D_{t+3}(\alpha)) \le L^{-2.5}\lambda(D_{t+3}(\alpha))$ for $t = t_n$. For fixed $t \ge k'$ the event $\{a_{k'+1} > T\}$ is a union of cylinders of depth $t+3$, so $\mathbb{P}\big(a_{k'+1} > T,\ \operatorname{dist}(\alpha, \partial D_{t+3}(\alpha)) \le \epsilon\lambda(D_{t+3}(\alpha))\big) = 2\epsilon\mathbb{P}(a_{k'+1} > T)$; summing over $t \le M$ and $k' \le M$ gives $O(M^2L^{-2.5}/T) \to 0$. (v) Assume (i)–(iv) fail. The range of the level-$(k'+1)$ cell is of type **F**, **P** or **Q** (Proposition 4.5) with parameter within $\operatorname{osc}_{\mathcal{C}_{k'-s}} \alpha_{k'} \le 4\Pi_{k'}^2/N_{k'-s-1} \le 4L^{-7.6}$ of $\alpha_{k'}$ (Lemma 4.7(b)), and $\alpha_{k'}, 1-\alpha_{k'} \ge (2(U+1))^{-1}$, so its length is $\ge (4(U+1))^{-1}$; by the formula $|d\alpha/dy_{k'}| \ge (4N_{k'}q_{k'}^2)^{-1}$ from the proof of Lemma 4.6, $|\mathcal{C}_{k'+1}| \ge (16(U+1)N_{k'}q_{k'}^2)^{-1} \ge (16(U+1)X)^{-1}$, whereas $\lambda(D_{t_f}) \le q_{t_f}^{-2} \le L^{-4}X^{-1}$. Thus (v) implies $\operatorname{dist}(\alpha, \partial\mathcal{C}_{k'+1}(\alpha)) \le 32L^{-2.9}|\mathcal{C}_{k'+1}(\alpha)|$. The level-$(k'+1)$ cells partition $(0,1)$ and $\{a_{k'+1} > T\}$ is a union of them, so this has probability $\le 64L^{-2.9}\mathbb{P}(a_{k'+1} > T)$; sum over k'. □

Lemma 9.4. *For each k', the k'-th term of (9.2) is at most* $C\mathbb{E}[\xi_{k'}\mathbf{1}_{\mathcal{E}_{k'}}] + C\,\mathbb{P}(\xi_{k'}=1)\big(T^{-1} + sL^{-2.7} + \vartheta_Q'^{r_h} + |\theta|UL^{-5}\big) + C|\theta|2^{-s/2}L^{-1}\mathbb{E}[\xi_{k'}\#\{k : \xi_k = 1\}]$.

Proof. Off $\mathcal{E}_{k'}$ we have $g_{k'} = 1$, $P_{\rm np} = 1$, $F_{\rm tr} = 1$ and the set defining t_n is non-empty; inserting the factor $g_{k'}\mathbf{1}\{\text{the set defining } t_n \text{ is non-empty}\}$ and replacing $P_{\rm np}$ and $F_{\rm tr}$ by 1 costs at most $C\mathbb{E}[\xi_{k'}\mathbf{1}_{\mathcal{E}_{k'}}]$. The non-emptiness indicator depends only on $q_{k'}, q_{k'+2}$, hence on $a_1,\dots,a_{k'+2}$, and is therefore constant on k'-regular level-$(k'-s)$ cells. Put $\Delta = \Theta_{k'} - \hat\Theta_{k'}$.

The past. On $\{g_{k'} = 1\}$ we have $J > k'+1$, so Lemma 4.7(a) applies on $\mathcal{C}_{k'-s}(\alpha)$. Each factor of P' with $\xi_k = 1$ involves y_k with $k \le k'-2s-3$ and the function m_k, which is constant on $\mathcal{C}_{k'-s}(\alpha)$; as $|m_k(t) - m_k(t')| \le 2|t-t'|$, $\operatorname{osc}_{\mathcal{C}_{k'-s}} \Theta_k \le 2|\theta|a_{k+1}\operatorname{osc} y_k/L \le C|\theta|2^{-s/2}/L$. Let P'' be the function obtained from P' by freezing every y_k ($k < k'-2s-2$) at its value at a fixed point of the level-$(k'-s)$ cell of α; then $|P''| \le 1$, P'' is constant on level-$(k'-s)$ cells, and $|P' - P''| \le C|\theta|2^{-s/2}L^{-1}\#\{k : \xi_k = 1\}$.

The near future. If (iv) fails, $\mathcal{C}_{k'-s}(\alpha)$ lies in a single cylinder of depth t_n+3; then t_n and $F_{\rm near}$ are constant on the cell (the factors B_k, $k \le t_n+1$, depend on a_{k+1} and w_k, i.e. on $a_1,\dots,a_{t_n+2}$), and so is the indicator of $(\text{iv})^c$. Thus $Y := P''\xi_{k'}g_{k'}\mathbf{1}\{\text{non-empty}\}\mathbf{1}_{(\text{iv})^c}F_{\rm near}$ is $\mathcal{H}_{k'-s}$-measurable and $|Y| \le 1$.

The far future. On $\{t_n = t\}$, $F_{\rm far} = \Phi_t$, where $\Phi_t = \prod_{k \ge t+w'} B_k$ is a functional of the extended chain from time $t+w'$ (it depends on $\zeta_{t+w'}$, through w_k, and on the subsequent pieces), bounded by 1 and independent of ω. If (v) fails, $D := D_{t_f}(\alpha) \subset \mathcal{C}_{k'+1}(\alpha)$; there $|dy_{k'}/d\alpha| \le 4N_{k'}q_{k'}^2 \le 4X$, so $\operatorname{osc}_D y_{k'} \le 4L^{-4}$ and $\operatorname{osc}_D \Delta \le 8|\theta|UL^{-5}$; moreover Y, t_n and $\mathbf{1}_{(v)^c}$ are constant on D (note $t_n+3 \le t_f$ and $D \subset \mathcal{C}_{k'-s}$). Since D is a cylinder of depth $t_f \le t_n + w' - r_h$ (9.3), Lemma 3.6(a) gives $\mathbb{E}[F_{\rm far} \mid D] = \phi_{t_n} + O(\vartheta_Q'^{r_h})$ with $\phi_t = \mathbb{E}\Phi_t$. Hence

$$\mathbb{E}\big[Y\Delta\mathbf{1}_{(v)^c}F_{\rm far}\big] = \mathbb{E}\big[Y\phi_{t_n}\Delta\mathbf{1}_{(v)^c}\big] + O\big(\mathbb{P}(\xi_{k'}=1)(\vartheta_Q'^{r_h} + |\theta|UL^{-5})\big).$$

Removing $\mathbf{1}_{(v)^c}$ costs $C\mathbb{E}[\xi_{k'}\mathbf{1}_{\mathcal{E}_{k'}}]$, and $Y\phi_{t_n}$ is $\mathcal{H}_{k'-s}$-measurable, so we are left with $\mathbb{E}\big[Y\phi_{t_n}\mathbb{E}[\Delta \mid \mathcal{H}_{k'-s}]\big]$. On $\{\xi_{k'}g_{k'} = 1\}$, $\Theta_{k'}$ is a function of $y_{k'}$ bounded by 2 ($m_{k'}$ being constant on the cell), so Theorem 5.3 and Lemma 5.2(ii) give, with $p = p_{k'}^{\mathcal{C}_{k'-s}} \le 2\alpha_{k'}$,

$$|\mathbb{E}[\Delta \mid \mathcal{H}_{k'-s}]| \le 4\big\|\operatorname{Law}(y_{k'} \mid \mathcal{C}_{k'-s}) - \nu_{\alpha_{k'},p}\big\|_{TV} + 4\big\|\nu_{\alpha_{k'},p} - \nu_{\alpha_{k'},\alpha_{k'}}\big\|_{TV} \le C(s+2)L^{-2.7} + 8\alpha_{k'},$$

with $\alpha_{k'} < 1/T$. □

Proof of Proposition 9.1. Sum Lemma 9.4 over k'. Using $\sum_{k'} \mathbb{P}(\xi_{k'} = 1) \le 2M/T \le 7/\varepsilon$, $\mathbb{E}[\#\{k : \xi_k = 1\}^2] \le C/\varepsilon^2$ and Lemma 9.3, the right-hand side of (9.2) is at most $C(\theta)\sqrt{\eta}/\varepsilon + o(1)$. With Lemma 9.2 this gives $\limsup_N |\mathbb{E}e^{i\theta H_2/L} - \varphi_\varepsilon(\theta)| \le C(\theta)\sqrt{\eta}/\varepsilon$, and (9.1) with $\eta \to 0$ gives $\mathbb{E}e^{i\theta H/L} \to \varphi_\varepsilon(\theta)$ for every θ. □

10. Proof of Theorem 1.1

By (7.3), $D_N(\alpha, \beta; r)/L = (\Lambda + \Lambda_Z)/L + H/L + R/L$. By Lemma 7.4, $R/L \to 0$ in probability. For fixed ε, $H/L \Rightarrow \Pi_\varepsilon$ (Proposition 9.1) and $\limsup_N \mathbb{P}(|\Lambda + \Lambda_Z|/L > \delta) \le C_Q\varepsilon/\delta^2$ (Proposition 8.1). Since $\int_0^\infty (1 - \cos cx)x^{-2}dx = \frac{\pi}{2}|c|$ and $\int_0^\varepsilon (1 - \cos cx)x^{-2}dx \le c^2\varepsilon/2$,

$$\lim_{\varepsilon \to 0} \log \varphi_\varepsilon(\theta) = -\frac{c_0}{\log 2} \cdot \frac{\pi|\theta|}{2} \cdot \mathbb{E} \int_0^1 |m(t, c_Q)|\, dt = -\frac{6}{\pi}\, \mathbb{E}F\big(\{z_1 c_Q\}, \{z_2 c_Q\}\big)\, |\theta| = -\sigma(\beta, r)|\theta|,$$

because $\frac{c_0}{\log 2} \cdot \frac{\pi}{2} = \frac{6}{\pi}$ and $\int_0^1 |m(t, c)|dt = F(\{P_1 c/Q\}, \{P_2 c/Q\})$ with $P_i/Q = z_i$. So Π_ε converges to the Cauchy law with characteristic function $e^{-\sigma(\beta,r)|\theta|}$. By the standard approximation theorem [1, Theorem 3.2], $D_N(\alpha, \beta; r)/\log N$ converges in distribution to this law when $\alpha \sim \lambda$. Its distribution function $\frac{1}{2} + \frac{1}{\pi}\arctan(c/\sigma(\beta, r))$ is continuous, which proves Theorem 1.1. □

Remark 10.1 (Where rationality is used)**.** Rationality of β and r enters only through Section 3: Lemma 3.1 (the endpoint orbits live on finitely many lines $z = \{u + v\alpha_k\}$), the Markov refinement at the finite set E_0, and the coupling Lemma 3.5. For irrational endpoints the same decomposition holds, and the proof would go through verbatim given a law of large numbers, with a rate $o(1/\log L)$ for the alternating version, for functionals of $(\omega_k, \{z_i q_k\})$; this is the equidistribution problem for the orbit $(\{z_i q_{k-1}\}, \{z_i q_k\})$ of the torus under the random products of $\left(\begin{smallmatrix} 0 & 1 \\ 1 & a \end{smallmatrix}\right)$ driven by the continued fraction digits, which we do not address. Borda raised the analogous question for the maximal sums [3].

Remark 10.2 (Symmetries)**.** For irrational α one checks $D_N(1 - \alpha, \beta; r) = D_N(\alpha, r - \beta; r)$ and $D_N(\alpha, r; r) = -D_N(\alpha, 0; 1 - r)$. For $\beta = 0$, $r = \frac{1}{2}$ this gives $D_N(1 - \alpha; \frac{1}{2}) = -D_N(\alpha; \frac{1}{2})$, so in that case the symmetry of the limit law is obvious. In general these identities relate different starting points, and the centring of the limit law comes from the parity-free dynamics of Section 3; consistently with them, $\sigma(\beta, r) = \sigma(r - \beta, r)$ and $\sigma(r, r) = \sigma(0, 1 - r)$, since $\{-\beta\}$ and $\{r - \beta\}$ are then exchanged or reflected.

11. Numerical checks

All computations use exact integer arithmetic. We take $\alpha = A/2^B$ with A uniform and $2^B \ge N^4$, evaluate $D_N(\alpha; z)$ by a Euclid-type floor-sum algorithm, and use $D_N(\alpha, \beta; r) = D_N(\alpha; z_2) - D_N(\alpha; z_1)$ (Lemma 2.5). Everything below is reproduced by the script `verify_kesten.py` accompanying this paper.

Exact identities. (a) The expansion of Proposition 2.4 was compared with the direct count for 600 random triples (α, z, N) (α with denominators 2^{96}–2^{256}, $z = P/Q$ with $Q \le 1000$, N up to 10^{14}): no mismatch. (b) The structure of Lemma 3.1 ($z_k = \{u_k + v_k\alpha_k\}$, $v_k \equiv P(w_k)_1/Q$, $-2 - 2\alpha_{k-1} < v_k \le 0$) was checked at 28000 levels ($Q \in \{2, 3, 5, 7, 12\}$): no violation; the largest $|v_k|$ observed was 2.33, and the reset $v_{k+1} = -u_k$ after a partial quotient $\ge 4Q + 1$ with $u_k \ne 0$ (Step 2 of Lemma 3.5) held in all 1786 instances. (c) For y uniform, $\int_0^1 \ell(x, y, z)\, dy = \frac{z(1-z)}{2x} + z' + \frac{z}{2} - 1 - \frac{xz'(1-z')}{2}$ with $z' = \{-z/x\}$; this was verified exactly for 300 rational pairs (x, z). It shows that the bounded part of the level function has a non-zero mean, depending on the endpoint orbit; its alternating sum is $o(L)$ only because even and odd levels have the same statistics (Section 3).

The limit law at rational parameters. For $N = 10^{30}$ and 20000 independent α per case we list quantiles of $X = D_N(\alpha,\beta;r)/\log N$ and the scale estimates $\hat\sigma(\theta) = -\theta^{-1}\log|\frac{1}{n}\sum\cos(\theta X)|$ (for a Cauchy law with scale σ, $\hat\sigma(\theta)\to\sigma$), next to the exact value of $\sigma(\beta,r)$:

(β,r)	$\pi\sigma(\beta,r)$	$\sigma(\beta,r)$	$q_{.10}$	$q_{.25}$	$q_{.75}$	$q_{.90}$	$\hat\sigma(1)$	$\hat\sigma(2)$
$(0,\frac12)$	$\frac12$	.1592	−.478	−.159	.159	.478	.1551	.1573
$(0,\frac13)$	$\frac12$	.1592	−.483	−.150	.154	.473	.1543	.1569
$(0,\frac27)$	$\frac12$	.1592	−.467	−.163	.155	.474	.1524	.1576
$(\frac14,\frac12)$	$\frac14$	.0796	−.246	−.087	.072	.232	.0782	.0777
$(\frac15,\frac25)$	$\frac14$	.0796	−.246	−.087	.087	.232	.0765	.0786
$(\frac23,\frac13)$	$\frac14$	.0796	−.236	−.077	.082	.256	.0744	.0785
$(\frac17,\frac37)$	$\frac{3}{10}$	.0955	−.296	−.108	.095	.298	.0941	.0965
$(\frac13,\frac12)$	$\frac13$	.1061	−.333	−.116	.116	.333	.1091	.1074
$(\frac27,\frac12)$	$\frac13$	.1061	−.318	−.116	.101	.318	.1004	.1035
$(\frac{7}{10},\frac{3}{10})$	$\frac{499}{1512}$	.1051	−.333	−.116	.116	.318	.1030	.1041
$(\frac12,\frac13)$	$\frac38$	.1194	−.352	−.121	.125	.357	.1129	.1163

For a Cauchy law with scale σ the quartiles are $\pm\sigma$ and the deciles $\pm3.078\,\sigma$. The quantiles are affected by the lattice structure of D_N (its values are spaced by $1/\log N \approx 0.0145$ here), the estimates $\hat\sigma(\theta)$ are not. The estimates $\hat\sigma(2)$ (standard error about 0.0035) agree with the predicted scales to within 1–3%, the noisier $\hat\sigma(1)$ (standard error about 0.006) to within one or two standard errors, and both separate the different values of $\sigma(\beta,r)$ clearly. A small downward bias of a few per cent was also observed for the sawtooth sums in [6]; it reflects the slow convergence (there are only about $0.84\log N \approx 58$ renormalisation steps).

For $N = 10^{60}$ (12000 samples per case) the estimates are:

(β,r)	$\sigma(\beta,r)$	$q_{.10}$	$q_{.25}$	$q_{.75}$	$q_{.90}$	$\hat\sigma(1)$	$\hat\sigma(2)$
$(0,\frac13)$	.1592	−.458	−.154	.157	.454	.1547	.1528
$(\frac14,\frac12)$	.0796	−.246	−.080	.080	.239	.0763	.0782
$(\frac17,\frac37)$	.0955	−.293	−.097	.098	.286	.0954	.0966
$(\frac{7}{10},\frac{3}{10})$	.1051	−.326	−.109	.109	.326	.1074	.1049
$(\frac12,\frac13)$	.1194	−.372	−.118	.128	.374	.1182	.1225

The standard error of $\hat\sigma(2)$ is about 0.003 for 20000 samples and 0.004 for 12000 samples, so all entries are compatible with the predicted scales.

Centring. The most delicate point of the proof is that the drift produced by the bounded terms is $o(\log N)$ (Section 8). We estimated the location $\hat c(\theta) = \theta^{-1}\arg\big(\frac{1}{n}\sum e^{i\theta X}\big)$ for two asymmetric pairs, with 40000 samples per value of N:

N	10^{12}	10^{20}	10^{45}	10^{60}
$(\frac17,\frac37)$: $\hat c(2)$	−.0094	−.0075	−.0021	−.0019
$\hat c(2)\log N$	−.26	−.35	−.21	−.27
$(\frac{7}{10},\frac{3}{10})$: $\hat c(2)$	.0052	.0020	.0028	.0016
$\hat c(2)\log N$	.15	.09	.29	.22

(standard error of $\hat c(2)$ about 0.0013). The location of D_N itself thus stays bounded, as predicted by the proof, and $\hat c\to 0$.

Irrational parameters (Conjecture 1.5). With the same N and sample size:

(β, r)	conjectured σ	$q_{.10}$	$q_{.25}$	$q_{.75}$	$q_{.90}$	$\hat{\sigma}(1)$	$\hat{\sigma}(2)$
$(0, \sqrt{2}-1)$	$\frac{1}{2\pi} = .1592$	$-.461$	$-.157$	.162	.466	.1516	.1548
$(0, 1/\pi)$	$\frac{1}{2\pi} = .1592$	$-.471$	$-.152$	.152	.470	.1538	.1551
$(0, \frac{\sqrt{5}-1}{2})$	$\frac{1}{2\pi} = .1592$	$-.473$	$-.155$	.149	.468	.1531	.1555
$(1/\pi, \sqrt{2}-1)$	$\frac{1}{3\pi} = .1061$	$-.330$	$-.113$	.104	.321	.1067	.1054
$(e-2, \frac{\sqrt{5}-1}{2})$	$\frac{1}{3\pi} = .1061$	$-.314$	$-.111$	.120	.323	.1001	.1018

(Here β and r are replaced by dyadic approximations with error below 2^{-400}, which does not affect D_N for $N = 10^{30}$ except on a set of negligible probability.)

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